ECEN215 LAB KIT 2

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Final Report

REVISION – 2
27 April 2026

Change Record

Rev.	Date	Originator	Approvals	Description
1	12/6/2025	ECEN215 Labkit		Final ECEN403 Release
2	4/27/2026	ECEN215 Labkit		Final ECEN404 Release

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ECEN215 Lab Kit 2

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Concept of Operations

REVISION – 3
27 April 2026

Concept of Operations
FOR
ECEN215 Labkit

Approved By:

Gavin Bernard 4/27/2026
Project Leader Date

Prof. Jang Date

Hailey Ward 4/27/2026
T/A Date

Change Record

Rev.	Date	Originator	Approvals	Description
-	9/17/2025	ECEN215 Labkit		Draft Release
1	10/04/2025	ECEN215 Labkit		Draft Revisions
2	12/06/2025	ECEN215 Labkit		Revisions
3	4/27/2026	ECEN215 Labkit		Final 404 Release

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1. Executive Summary

In the ECEN215 course, Principles of Electrical Engineering, students have been completing online labs in their electrical engineering classes by watching a professor perform the lab himself, then writing their own report. Because these students aren't physically doing the labs, they aren't able to gather necessary experience working with electrical engineering equipment. In order to promote more hands-on learning, creating an affordable and accessible system, such as the ECEN215 Labkit, will allow for further education within the course.

2. Introduction

ECEN 215, Principles of Electrical Engineering, is a lab-based course meant for non-electrical engineering students. The ECEN 215 Labkit will serve as a portable and self-contained system, where students can complete their laboratory assignments without access to the on-campus lab. The objective of this project is to increase accessibility to the course in an affordable manner.

2.1 Background

The ECEN215 Labkit is a self-contained system for use by non-electrical engineering students that do not have access to on-campus or professional laboratory equipment. This system will contain a multimeter, power supply, oscilloscope, and signal generator capable of the requirements of each lab, accompanied by a mobile application that allows students to interface with the device. Students will use the device to take all measurements required by the lab, supply inputs to their circuits, and interface with the data using the mobile app.

2.2 Overview

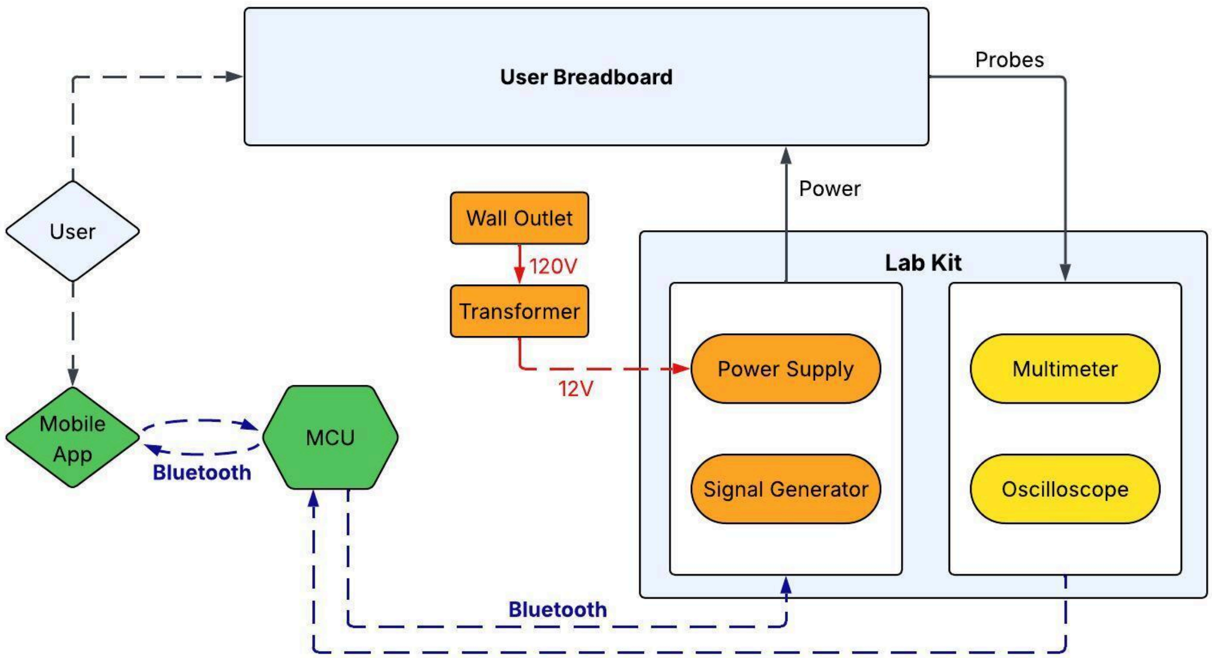

Figure 1: System Description Block Diagram

Our system will be a small, portable, and cost-effective electronic testing device that combines multiple functions, including a multimeter, oscilloscope, signal generator, and power supply. As seen in **Figure 1**, the inputs, outputs, and users must interact back and forth with the microcontroller unit, as the user, through the mobile app, selects what inputs to apply and what outputs to measure. Between the inputs and the outputs is the users' breadboard, which serves as the device under test and connection point for input and output probes. External connections include a wall outlet for the input, in order to supply power to the user and device.

2.3 Referenced Documents and Standards

- IEEE 802.15 Working Group for Wireless Personal Area Networks (WPANs)
- IEEE 1241-2010 Terminology and Test Methods for Analog-to-Digital Converters
- IEEE 1057-2017 Digitizing Waveform Recorders
- IEEE 1139-2022 Standard Definitions of Physical Quantities for Fundamental Frequency and Time Metrology
- IEEE 946-2020 Recommended Practice for the Design of DC Power Systems for Stationary Applications
- IEEE 1100-2005 Recommended Practice for Powering and Grounding Electronic Equipment
- W. Trinh, J. Tyler, S. Villareal, M. Rahmian, and N. Gober, "ECEN 215 Lab Manual"

3. Operating Concept

3.1 Scope

The ECEN215 Labkit will combine hardware and software into an integrated system, consisting of the physical measurement device and mobile application. The device will be able to generate signals, supply constant voltages, and take electrical measurements from the student's circuits. These measurements will be transmitted through BLE to the mobile application, which serves as the user interface for viewing the data, as well as controlling the output signals and voltages. This kit will replicate the functionality of traditional lab instruments, allowing students to perform ECEN215 lab experiments without accessing on-campus lab facilities.

3.2 Operational Description and Constraints

The ECEN215 Labkit is intended for use by non-electrical engineering students outside of a laboratory setting. The student builds the circuit required by the lab, then utilizes the Labkit system in order to supply power, generate signals, and display waveforms and measurements through the mobile app. Possible labs include simple circuit building, amplifier circuits, AC signal circuits, and logic gates.

The operational constraints from this description are as follows:

- Compatible with devices running Android 16 or newer
- Input under 5V and 10kHz
- Powered by a standard US wall outlet
- Capable of completing all labs detailed in the ECEN 215 Lab Manual

3.3 System Description

The ECEN215 Labkit can be broken down into 5 subsystems, detailed below:

- Power Supply: The system's power supply subsystem will regulate power from a standard US wall outlet to supply power to all hardware, and output -5V and constant voltages between 0 and 5V to the user.
- Signal Generation: The signal generator will generate signals between 20Hz and 1.5kHz, with amplitudes between 0.5 and 2V, as well as sine, square, and ramp waveforms.
- Multimeter: The multimeter will measure voltages between -5 and 5V, resistances between 10Ω and 10kΩ, and well as currents between 0 and 1A.
- Oscilloscope: The oscilloscope will sample signals at 10kHz with two channels.
- Mobile Application: The mobile app allows the user to control inputs such as voltage and signals, and view outputs such as waveforms and voltage readings by communicating with the MCU. In addition, the app will provide the user with lab information, such as the manuals and connection information.

3.4 Modes of Operations

- The ECEN215 Labkit is made to specifically run the labs from the ECEN 215 lab manual. The Labkit will only be capable of completing those labs in particular.

3.5 Users

The specific users for design will be students in the ECEN215 course, who are non-electrical engineering students, and possibly creating a circuit for the first time. They will be outside of a laboratory setting and won't have direct access to a Teaching Assistant that can help troubleshoot their circuit. In addition, they may not completely understand what each function of the device is able to do, and how it works.

3.6 Support

Support for the ECEN215 Labkit will be in the form of a user manual outlining the mobile application installation, system usage, and maintenance, available on the mobile app. Direct support will be available by contacting Texas A&M Professor John Lusher.

4. Scenarios

The ECEN215 Labkit is designed to be used for coursework or for basic freestyle measurements.

4.1 ECEN 215 Labs

Our main scenario is for students taking the ECEN 215 course. The device is specifically designed to work for labs 1 through 8, and will walk the student through each lab step-by-step, automatically providing the correct outputs, and will indicate values to measure. The data for each lab will be displayed on the mobile app throughout the progress of the lab and be available for export upon request of the user.

5. Analysis

5.1 Summary of Proposed Improvements

The ECEN215 Labkit is relatively affordable compared to the current device solution used in the course. The Labkit's mobile application and system are designed for the ECEN215 course, so the system features are specific to the requirements of the assignments. Limiting additional features will further reduce confusion for students. In addition, the system includes everything needed to complete the labs, eliminating the need for students to purchase additional equipment or technology.

5.2 Disadvantages and Limitations

The ECEN 215 Labkit will have some limitations compared to the current device solution used. Designed to be low cost, the Labkit won't contain as many features or be compatible with as wide of a range of supply or measurements as the current lab solution. In addition, real time analysis of signals will not be possible using the app. Instead the user will analyze the raw graphed data captured using the app. Compared to the current solution, the Labkit will be powered by a standard wall outlet and controlled using a mobile phone, as opposed to a laptop being the power source and interface method currently used.

5.3 Alternatives

With the current issue of the course being students not completing the lab themselves, an alternative and the most accessible solution would be online simulations similar to each lab that allow students to take their own measurements and build circuits. In addition, renting a Labkit or lab equipment from a nearby university or online retailer will allow students to take physical measurements, but likely comes with financial burden. The least accessible alternative would be to commute to campus to complete the lab, but would allow the student to perform the lab as originally designed.

5.4 Impact

The impact of the Labkit would be most significant towards the students enrolled in the ECEN 215 course. They would be able to experience and learn the course content in more depth by having hands-on work instead of watching an online lab. Outside of the coursework, students would be able to pursue hobby circuit projects with the free measuring mode, as noted in Scenario 4.2 Free Measuring Mode, allowing them to further explore electrical engineering. Professors would also benefit from students' increased understanding of course content, which likely leads to higher assignment and test scores. On top of this, the Labkit would help the Electrical and Computer Engineering department by creating an affordable solution to the current device used.

ECEN215 Lab Kit 2

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Functional System Requirements

REVISION – 2
27 April 2026

Functional System Requirements
FOR
ECEN215 Labkit

Approved By:

<u>Gavin Bernard 4/27/2026</u>
Project Leader Date

____
Prof. Jang Date

<u>Hailey Ward 4/27/2026</u>
T/A Date

Change Record

Rev.	Date	Originator	Approvals	Description
-	10/04/2025	ECEN215 Labkit		Draft Release
1	12/06/2025	ECEN215 Labkit		Revisions
2	4/27/2026	ECEN215 Labkit		Final 404 Release

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1. Introduction

1.1 Purpose and Scope

Allowing students to complete ECEN 215 labs remotely is important for future students' understanding of electrical engineering. Rather than having students spend an excessive amount on lab materials, it is important to have a system that is relatively cheap so each student can complete each lab at an affordable cost. The ECEN215 Labkit will be able to compute all different aspects of each lab, have affordable-priced components, and be small enough that it is easily portable.

1.2 Responsibility and Change Authority

The team leader, Gavin Bernard, will be responsible for making sure that each team member can complete each subsystem on time with technical merit. These requirements can only be changed with the approval of the team leader and Professor Stavros Kalafatis or Professor John Lusher.

Subsystem	Responsibility
Multimeter	Maylun Huang
Oscilloscope	Maylun Huang
Mobile App	Evan Slyter
Signal Generator	Gavin Bernard
Power Supply	Gavin Bernard

Table 1. Subsystem Responsibility

2. Applicable and Reference Documents

2.1 Applicable Documents

Document Number	Revision/Release Date	Document Title
IEC 61010-1	February, 2020	Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 1: General requirements
IEEE 1057-2017	November, 2017	IEEE Standard for Digitizing Waveform Recorders
ISO/IEC/IEEE	June, 2017	Systems and Software Engineering — Software Life Cycle Processes

Table 2. Applicable Documents

2.2 Reference Documents

Document Number	Revision/Release Date	Document Title
IPC-2221	May, 2012	Generic Standard on Printed Board Design
IPC-2223	January, 2020	Sectional Design Standard for Flexible/ Rigid-Flexible Printed Boards
IPC-A-600	October, 2020	Acceptability of Printed Circuit Boards

Table 3. Reference Documents

2.3 Order of Precedence

In the event of a conflict between the text of this specification and an applicable document cited herein, the text of this specification takes precedence without any exceptions.

All specifications, standards, exhibits, drawings or other documents that are invoked as “applicable” in this specification are incorporated as cited. All documents that are referred to within an applicable document are for guidance and information only, apart from ICD’s that have their applicable documents considered to be incorporated as cited.

3. Requirements

3.1 System Definition

The ECEN 215 Labkit is a multi-use device to assist users in completing 8 pre-written labs designed to teach basic electrical laws, techniques, and devices. It will let users complete these labs without having to purchase expensive equipment and without having access to a lab. The Labkit has 5 subsystems: Multimeter, Oscilloscope, Signal Generator, Power Supply, and an App to interface with the device.

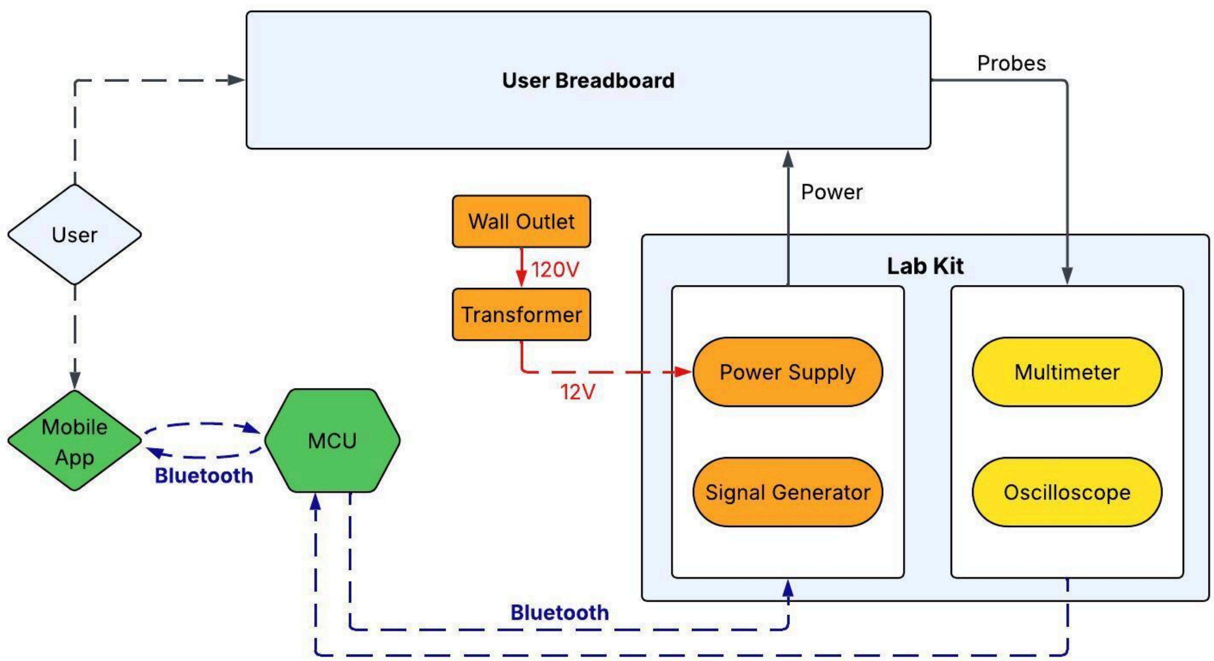

Figure 1. Detailed System Diagram

The Labkit will be powered by a direct connection to a wall outlet; no batteries will be used except for the mobile device interfacing with the Labkit. The connection used to power the device will be used for DC power output as well as AC signal generation.

The configuration of the specific subsystems will be automatically controlled by the app. Measurement devices will be activated as needed based on which step of the lab manual the user is working on. Signal generation and DC power output will be activated similarly to ensure the user is unable to accidentally output dangerous amounts of current or begin outputting current before the probes are where the user decides to put them.

3.2 Characteristics

3.2.1 Functional / Performance Requirements

3.2.1.1 Alternating Current to Direct Current

The current within the printed circuit board shall have a direct current for all the labs within ECEN 215. In addition, the components on the board need to have a specific power supply to operate. The voltage shall be directed from a 120-volt AC source down to a 12-volt output.

Rationale: Without the specific current, the components won't be able to operate and control the different subsystems. The MCU needs to be able to control these, and without the power supply, it won't be able to. As well, we need to be able to output variable voltage, which will come directly from the power supply.

3.2.1.2 Regulating Voltage

The system shall regulate voltage from 12 volts to a 3.3 volt rail, a -5 volt rail, and a 5 volt rail.

Rationale: The high voltage entering the PCB could either hurt a student or destroy the entire subsystem and prevent them from continuing with the lab. By having regulators in place, we can control the voltage and limit damage to either the student or the subsystem.

3.2.1.3 Sweeping Voltage

The MCU shall control the voltage output and be able to sweep from 0 volts to 5 volts.

Rationale: Within each lab manual, the voltage needs to be able to be controlled as an output for each circuit. Most labs have the voltage output either being -5 volts or 5 volts, but there are some requirements for it to be within that range, making a need for a variable voltage.

3.2.1.4 Variable Frequency

The frequency within each lab needs to be different, controlled, and varied, as all labs require a range of 20 Hz to 1,500 Hz.

Rationale: Each lab manual requires the student to be able to control the frequency needed in the lab manual. As well, the frequency output needs to have minimal noise to allow the students to read specific results within the app.

3.2.1.5 Multiple Wave Signals

The signal generator shall output 3 different types of waves, such as sinusoidal, square, and triangle.

Rationale: Students need to be able to understand how each wave type works and how the results differ from each other. Each lab manual requires frequency use of a specific type of wave, so the student needs to be able to control the output type to receive the correct measurements.

3.2.1.6 Phone Compatibility

The interface app shall be usable on Android devices running Android 8.0 or newer.

Rationale: Older Android devices do not support current software updates that are needed for security. Older devices also do not have the required Bluetooth capabilities needed to send and receive information from the Labkit. Since this app only supports Android devices, the user must not be using an IOS device.

3.2.1.7 Reading Voltage, Resistance, and Current

The Multimeter shall take voltage, resistance and current measurements within 5% error and transmit them to the MCU.

Rationale: In order to meet the lab requirements, students must take accurate measurements of their system, and access their data.

<u>3.2.1.8 Oscilloscope Data</u>

The oscilloscope shall record voltages with respect to time, display data in graph form, with the ability to scrub through data, with single and dual channels.

Rationale: In order to meet the lab requirements, students must be able to record oscilloscope data and manipulate it in order to find specific data points.

3.2.2 Physical Characteristics

<u>3.2.2.1 Mass</u>

The mass of our entire system is going to be relatively light, under 300g.

Rationale: Texas A&M University would have to ship these Labkits to each student that is virtual. To reduce the cost as much as possible, the lighter the system, the more it saves on shipping costs. As well, the system needs to be portable, so it has to be light enough for each student to be able to travel with it.

<u>3.2.2.2 Volume</u>

The entire system needs to be portable, so the volume of our entire system will be around 1125m^3, or 0.15m x 0.15m x 0.05m.

Rationale: One of the system's main aspects is that the system needs to be portable. So the entire box encasing the system needs to be small enough for students to carry or fit in their backpacks.

3.2.3 Electrical Characteristics

<u>3.2.3.1 Inputs</u>

3.2.3.1.1 Input Voltage Level

The input voltage for the ECEN 215 Labkit shall be 3.3 volts and 5 volts.

Rationale: The needed voltage for the MCU, Regulators, ADC, and DAC is either 3.3 volts or 5 volts. To ensure that any of the subsystems function properly, we require the power to reach these components quickly. Additionally, we need to be able to output a voltage range of -5 volts to 5 volts for the students' breadboard.

3.2.3.1.2 Power Consumption

The maximum peak power shall not exceed 2 Watts.

Rationale: The peak current required is 0.405 Amps, and a maximum of 5 Volts is going into the circuit. So the maximum power consumed mainly by the ESP32 is roughly 1 Watt, and the other Digital to Analog Converters and Buck converters can consume up to 1 Watt of power.

<u>3.2.3.2 Outputs</u>

3.2.3.2.1 Data Output

All data output from our system will be uploaded and viewed from our app design.

Rationale: Every student has a phone that they can download an app and use to connect to our system. This makes our system accessible to everyone for free and allows them to go through the lab manuals directly on the app.

3.2.3.2.2 Oscilloscope and Multimeter Output

Within our system, our oscilloscope and multimeter shall be able to output data to our MCU.

Rationale: With these two subsystems directly connecting data to our MCU, it allows us to send the data to the user. This will make the entire system more user-friendly and allow for a seamless system.

3.2.3.3 Connectors

The ECEN 215 Labkit shall follow the guidelines of the Association Connecting Electronics Industries, IPC-2221.

Rationale: Conform to industry standards

3.2.3.4 Wiring

The ECEN 215 Labkit shall follow the guidelines of the Association Connecting Electronics Industries, IPC-2223.

Rationale: Conform to industry standards

3.2.4 Environmental Requirements

The environment for our ECEN 215 Labkit will need to function inside or outside.

Rationale: The Labkit needs to be portable so students can work on their labs in different locations which could be outside or inside

3.2.4.1 Sealed Container

Our ECEN 215 Labkit shall be in a secure container that allows for transportation.

Rationale: The system needs to be able to withstand being transported from locations so the student can work on it from home or on campus.

3.2.5 Failure Propagation

The ECEN 215 Labkit shall not allow fault propagation throughout the system past the generic use of the essential equipment

3.2.5.1 Data Isolation

The ECEN 215 Labkit shall save the data in between uses in case of short or kit failure

Rationale: Once a student records data after measuring an aspect of their circuit, the system will save that data and pause until the next use. This allows the student to take measurements and not lose their data.

4. Support Requirements

The ECEN 215 Labkit requires a mobile device that has a stable internet connection to export the data from the labs. The mobile device must also be new enough (specifics stated above) that it can support the app and receive/send BLE signals. The system requires a wall outlet to supply supporting power for all components within the system. The Labkit has (1) MCU, (2) DACs, (2) Buck converters, and (2) ADCs.

ECEN215 Lab Kit 2
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Interface Control Document

REVISION – 2
27 April 2026

Interface Control Document
FOR
ECEN215 Labkit

Approved By:

Gavin Bernard 4/27/2026
Project Leader Date

Prof. Jang Date

Hailey Ward 4/27/2026
T/A Date

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-	10/04/2025	ECEN215 Labkit		Draft Release
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1. Overview

This document details how the ECEN215 Labkit's components, the multimeter, power supply, oscilloscope, signal generator, and mobile app will interface. The ICD will include physical descriptions of all the components, including electrical, power, and thermal information of the different elements. In addition, this document will detail how the different subsystems will interact with each other in order to meet the system requirements.

2. References and Definitions

2.1 References

Document Number	Revision/Release Date	Document Title
IPC-2221	May, 2012	Generic Standard on Printed Board Design
IPC-2223	January, 2020	Sectional Design Standard for Flexible/ Rigid-Flexible Printed Boards
IPC-A-600	October, 2020	Acceptability of Printed Circuit Boards

Table 1. Reference Documents

2.2 Definitions

MCU	Microcontroller Unit
DAC	Digital to Analog Converter
BLE	Bluetooth Low Energy

3. Physical Interface

3.1 Weight

3.1.1 Main Control Unit

Component	Weight [oz]	Quantity
MCP16301	0.00127	4
ESP32	0.65	1
TLV900x	0.003076	1
CD4053	0.034	1
CD4066	0.00988	1
ADS8321	0.00092	1
OPA705	0.0186	7

Table 2. Weight Specifications

3.2 Dimensions

3.2.1 Dimensions of System Components

Component	Length [mm]	Width [mm]	Height [mm]
MCP16301	3.1	1.8	1.3
ESP32	18	25.5	3.1
TLV900x	8.12	2.9	2.8
CD4053	19.3	6.35	4.57
CD4066	8.65	3.91	1.58
ADS8321	3	3	0.97
OPA705	2.9	1.6	1.45

Table 3. Dimensions of Labkit Components

3.2.2 Dimensions and Weight of Whole System

Component	Length	Width	Height	Weight
Enclosed System	11.1 Inches	9.45 Inches	1.89 Inches	4 Ounces

Table 4. Dimensions and Weight of System

3.3 Mounting Locations

3.3.1 Placement of System

The system will be placed on a flat surface with access to a standard wall socket.

3.3.2 Mounting of Microcontroller
The microcontroller and printed circuit board will be placed in an enclosed housing, with access to the I/O ports and cabling for the power supply.

4. Thermal Interface

The ECEN215 Labkit will not need any external thermal interfacing, as the ESP32 microcontroller is low power and will perform standard measurement and generation tasks. For the other system components, they will also not need thermal interfacing, as they will not be handling significant loads, and they draw minimal power.

5. Electrical Interface

5.1 Primary Input Power

The MCU will be powered with 3.3V provided from the power supply. The power supply will receive 12V from the external transformer and will split the voltage and current as needed for each component.

5.2 Voltage and Current Levels

Component	Voltage [V]	Current [A]	Power [W]
ESP32	3.3	0.405	1.34
External Transformer	120/12	5	60
MCP16301	12/5	0.305	3.66
Signal Generator	5	0.035	0.175
DAC	5	0.01	0.05
ADS8321E/250	5.25	0.0011	0.0085

Table 5. Maximum Voltage and Current Levels

5.3 Signal Interfaces

5.3.1 BLE Signal Interface
The ESP32 will communicate with the app using the built in 2.4 GHz BLE antenna and radio. Signals will be received by the user’s personal smartphone. Data will be sent in the compact binary format.

5.4 User Control Interface

The user control interface is a mobile application that communicates with the microcontroller unit. Data collection, power, and signal output will be automatically controlled by the app depending on what part of which lab the user is working on.

6. Communications / Device Interface Protocols

6.1 Wireless Communications

6.1.1 Bluetooth Low Energy
The microcontroller has an integrated Bluetooth Low Energy module. This connection enables communication between the user and the Labkit, and the ability to collect data and output voltages through the probes.

6.1.2 Compact Binary
The format for data sent from the Labkit to the app through the Bluetooth Low Energy signal is compact binary. This format allows for very fast transfers of simple numerical data. We will be using fixed field sizes as well as little-endian format to ensure signals are uniform and simple to understand using the app.

6.1.3 UTF-8
The format for data sent from the app to the Labkit is UTF-8 encoded Ascii commands. This format allows for simple commands to be easily sent in text form to the Labkit.

6.2 Microcontroller Input and Output

The microcontroller has digital and analog input and output pins that can read 0V-3.3V. Voltages outside of this range will be read through a voltage divider.

ECEN215 LAB KIT 2

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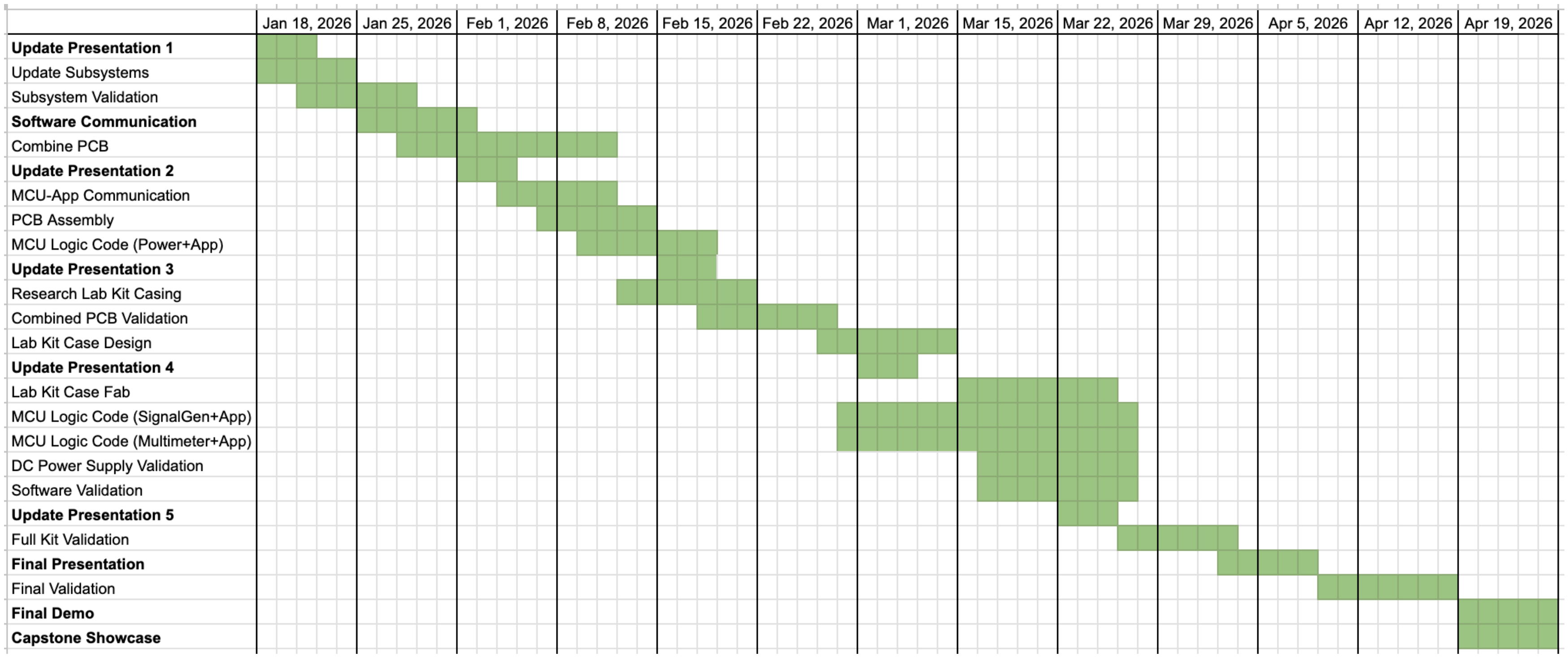
Schedule

REVISION – 2
27 April 2026

Execution Plan for ECEN215 Lab Kit



ECEN215 LAB KIT

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Evan Slyter

Validation

REVISION – 2
6 December 2025

Validation

Test Name	Success Criteria	Methodology	Status	Owner
Communication Range	Labkit connects to User's device within 8 ft	Connecting device to Labkit at 2ft intervals starting with device on top of Labkit.	VALIDATED	Evan
Outgoing Communication Time	Labkit sends information to device in <0.5s	A stopwatch will be started at the same time as the Labkit sends data to the phone. Once data is being received, the timer will be stopped.	UNTESTED	Evan
Connection Time	Labkit connects to users device within 30 seconds of scanning	A stopwatch will be started as soon as the device begins scanning for a bluetooth connection. Once the scan finds the Labkit and the Labkit successfully connects, the timer will be stopped	VALIDATED	Evan
Communication Accuracy	Binary data received on Labkit is the same data sent from User's device	Data is sent from the User's device to the Labkit. We will ensure we know which sequence is being sent and check what arrives at the MCU.	VALIDATED	Evan
Communication Strength	Labkit connects to User's device through minor obstacles.	Connecting the device to the Labkit when the Labkit is covered by a blanket, under a desk, and behind a door.	VALIDATED	Evan
Incoming Communication Time	Labkit receives information from device in <0.75s	A stopwatch will be started at the same time as the phone sends a command to the Labkit. Once the command is received and completed, the timer will be stopped.	VALIDATED	Evan
Power supplies correct voltage following command from app	Power supply successfully sets output voltage to whatever is said by binary data.	Once binary accuracy is confirmed, see if MCU code reads it correctly and sets output voltage as expected for voltages of 0,0.5,3.3, and 5.	VALIDATED	Evan + Gavin

Signal Generator can be controlled by the App	Signal Generator successfully changes the frequency,waveform, and voltage to whatever is inputted from the app.	Input from the app using code that was developed prior and then test using an oscilloscope if the output is correct to the expected input.	PARTIALLY VALIDATED	Evan + Gavin
Multimeter Voltage Testing	Measures -5 to 5V within 2% accuracy	Use bench multimeter to measure scaled voltage measurements, and compare to ideal measurements, then decode using MCU code	VALIDATED	Maylun
Multimeter Current Testing	Measures 0 to 1A within 2% accuracy	Use bench multimeter to measure scaled voltage measurements, and compare to ideal measurements, then decode using MCU code	VALIDATED	Maylun
Multimeter Resistance Testing	Measures 10 to 10kΩ within 2% accuracy	Use bench multimeter to measure scaled voltage measurements, and compare to ideal measurements, then decode using MCU code	VALIDATED	Maylun
Oscilloscope	Graph signals up to 10kHz	Use bench oscilloscope to graph outputted signal, and use MCU to graph output for single and dual channel	VALIDATED	Maylun
LabKit Measurements	LabKit receives and formats measurements	Apply multimeter and oscilloscope, and view output on Mobile App	PARTIALLY VALIDATED	Maylun
LabKit 5 Volt Output	LabKit is able to output 5 Volts and 1 Amps	12 Volt and 5 amp input uses a buck converter to step down the voltage and current to 5 volts and 1 amps	VALIDATED	Gavin
LabKit 3.3 Volt Output	LabKit is able to output 3.3 Volts and 1 Amp	Takes input and steps down the voltage to 3.3 volts and 1 amp to help power the ESP32 and other components within the system	VALIDATED	Gavin
LabKit Controllable Variable Voltage	LabKit is able to control a variable voltage output from 0 to 5 Volts	Uses a digital-to-analog converter that is connected to the ESP 32 that allows for a controllable output voltage that can be used for different aspects in the circuits	VALIDATED	Gavin
LabKit -5 Volt Output	LabKit is able to output -5 Volts with 40 mA	Uses an inverter to change the 5 volt input to -5 volts so	VALIDATED	Gavin

		students can operate op amps		
Sine, Ramp, and Square Signal Generator	Able to generate a Sine, Ramp, and Square Signal	Signal generator is connected to the ESP 32 that allows for a controllable waveform	PARTIALLY VALIDATED	Gavin
Signal Generation	Able to output a signal that can be measured	A waveform is able to be outputted and connected to a circuit for students to measure	VALIDATED	Gavin
20 - 1,500 Hz Frequency Change	Able to change the signal frequency in the range of 20 Hz to 1,500 Hz	With the Signal generator connected to the ESP 32, it can also change the frequency within the range expressed	PARTIALLY VALIDATED	Gavin
DC Power Supply Current Load	Able to power the entire product and power every lab needed.	While fully integrated, able to support every lab with enough current for students to take measurements.	VALIDATED	Gavin
System Reset Time	Labkit becomes available to connect to users device within 30 seconds of being powered	A stopwatch will be started as the Labkit is connected to power. The stopwatch will be stopped once the Labkit is detected by the user's device	VALIDATED	Evan
0.5 - 2.0 V Op Amp Controllable Output	Able to control the signal voltage from 0.5 Volts to 2 Volts	External op amp connected to a digital potentiometer can be regulated by the ESP 32 to help change the voltage of the signal created	PARTIALLY VALIDATED	Gavin

Performance on Execution Plan

The execution plan was very accurate, and we completed everything that was set forth at the beginning of the year. In March we did get behind on the PCB order, which set us back a week, but we soon caught up, where we made up the time in assembly. Also, some tasks were performed out of schedule to help speed up some of the process. Overall, everything was accomplished; the tasks set ahead at the beginning of the year, we completed in full.

Performance on Validation Plan

The validation plan for the lab kit was executed nearly entirely, with each individual subsystem undergoing thorough testing. Quantitative data was collected for power supply, signal generation, multimeter, and the oscilloscope subsystems, and functionality was confirmed for the mobile app subsystem. While several challenges were encountered during testing, specifically with validation and error percentages, all subsystems demonstrated stable operation. The experimental results confirm that the subsystems are functioning as intended and are suitable for full system integration.

ECEN215 LAB KIT 2

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Maylun Huang
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Subsystem Reports

REVISION – 3
27 April 2026

Subsystem Reports
FOR
ECEN215 Lab Kit

Approved By:

Gavin Bernard 4/27/26
Project Leader Date

Prof. Jang Date

Hailey Ward 9/26/26
T/A Date

Change Record

Rev.	Date	Originator	Approvals	Description
1	12/06/2025	ECEN215 Lab Kit		Draft Release
2	4/26/2026	ECEN215 Lab Kit		Updated for Final System
3	4/27/2026	ECEN215 Lab Kit		Final 404 Release

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1. Introduction

The ECEN215 Lab Kit system will integrate hardware and software in order to create an intuitive system for its users. The hardware subsystems, Power Supply, Signal Generation, Multimeter, and Oscilloscope, provide the tools to complete the lab, and the software subsystem, Mobile App, creates the framework for all interactions between the user and the device. With each subsystem validated separately to meet performance requirements, the integration results in a cohesive and user-friendly system that supports efficient lab interaction.

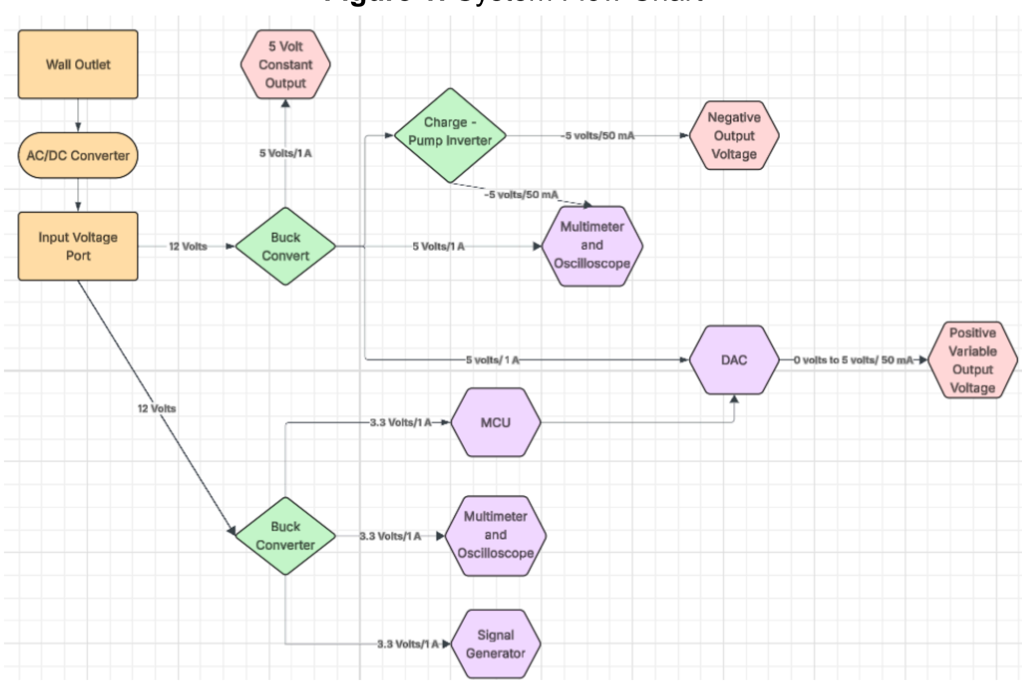
2. Power Supply Subsystem Report

2.1 Subsystem Introduction

As the entire goal of our system is to allow students to design circuits and complete measurement results. It is critical to have a constant voltage and current supply to power each aspect of the entire system. It is needed to have a 5 volt rail, a 3.3 volt rail, -5 volt output, and a variable voltage output to be able to complete every lab. As well, these rails needed to be able to power the entire system such as the multimeter, oscilloscope, signal generator, and microcontroller.

2.2 Subsystem Design

Figure 1: System Flow Chart



2.3 12 Volt to 5 Volt Output

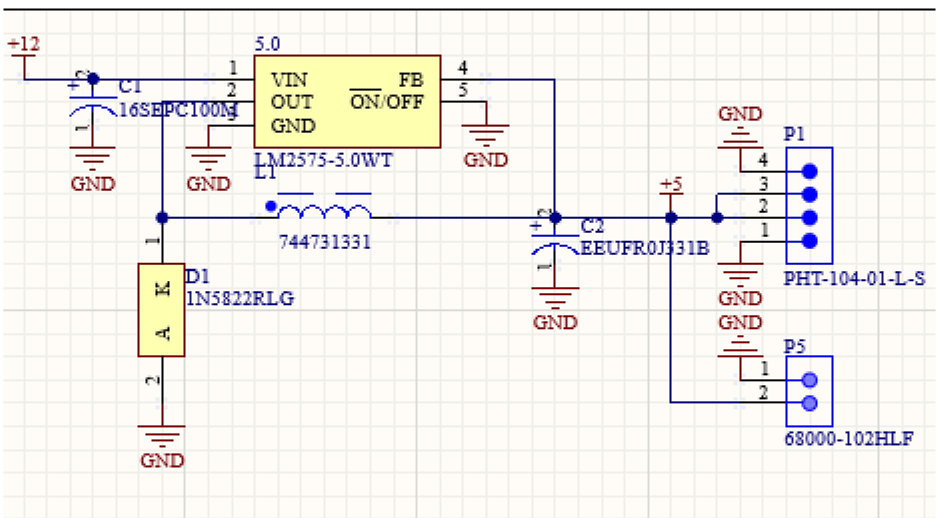
2.3.1. Operation

For the 5-volt output, it operates through a buck converter to decrease the voltage and increase the current for multiple devices in the student's circuit. It uses an LM2575-5.0WT chip integrated with only one capacitor, inductor, and schottky diode. The schematic design is directly from the LM2575-5.0WT data sheet as it gives a great example of how to use the circuity to drop the 12 volt input to a 5 volt output. The max output current is 1 amp which is plenty as most labs use large resistors and all chips within the product are low power.

2.3.2. Schematic Design

For this subsystem, the circuit involves 4 different components. The first component would be the schottky diode as it has very little voltage drop across it and the schottky diode I used was rated for 3 amps as I wanted to protect the circuit as much as possible as the current from the wall could spike. On top of that, I used a capacitor and inductor of both 330 microfarads which helps limit the voltage ripple on the output of the circuit. Finally, I also used a small filtering capacitor at the front of the buck converter which allowed for the input voltage to be as smooth as possible.

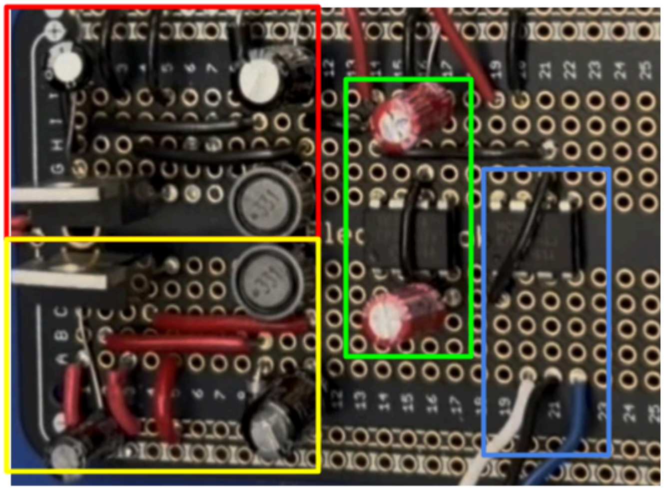
Figure 2: LM2575-5.0WT Schematic Design



2.3.3. Prototype

Before ordering the PCB of the DC power supply subsystem, I built the designed schematic above on a breakout board which allowed me to completely build the circuit and test it thoroughly. I was able to do this as every part was through-hole and had the capability to be tested on a breakout board. Within the red square on Figure 3, the 12 volt to 5 volt buck converter was built and made as compacted as possible to make it accurate with the PCB design.

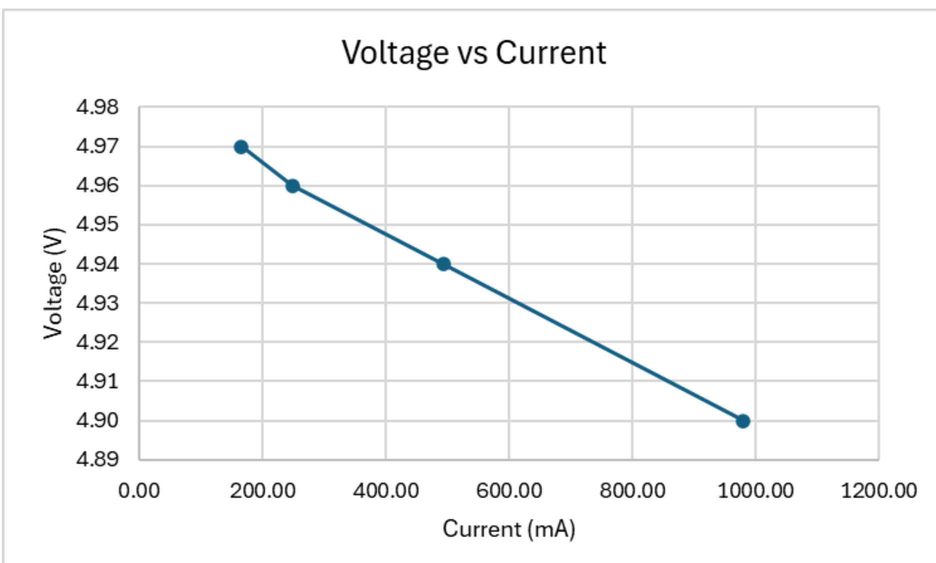
Figure 3: Prototype Buck Converter and Inverter Board



2.3.4. Validation

With the 12 Volt to 5 volt buck converter the main testing circled around the possible current load it could handle without any voltage sag. As explained before the max output current expected from the buck converter was 1 amp. As shown below, the circuit barely sagged as the output current increased which held steady for hours.

Figure 4: Current Load Testing of 12 Volt to 5 Volt Buck Converter



As shown in Figure 4, the voltage does sag as the current increases, but it only drops to 4.9 volts which is only a 2% error from the expected value of 5 volts. For the needed labs and chips within the system, 4.9 volts would be satisfactory for the needed requirements. As well, the current load rarely exceeds 500 mA for most of the labs and measurement tools.

2.4 12 Volt to 3.3 Volt Output

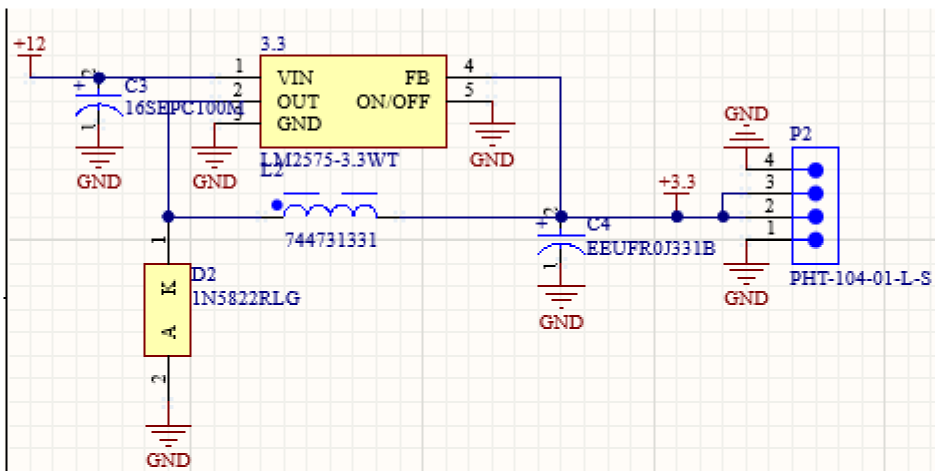
2.4.1. Operation

For the 3.3-volt output, the purpose is to power the microcontroller unit so that the system can be fully synchronized and controlled by the mobile app. The LM2575-3.3WT is the chip used within the system and the circuit is taken directly from the LM2575-3.3WT data sheet. This circuit uses the same values from the LM2575-5.0WT such as a schottky diode, inductor, and capacitors. As well, the rated current output of the buck converter is 1 amp.

2.4.2. Schematic Design

The schematic design for the LM2575-3.3WT is identical to the LM2575-5.0WT as they are sister chips. The circuit uses the same 330 microfarad capacitor and inductor to help smooth the output voltage ripple. As well, it uses a 3 amp schottky diode and a filter capacitor in the beginning of the circuit. This can all be seen below shown in Figure 5

Figure 5: LM2575-3.3WT Schematic



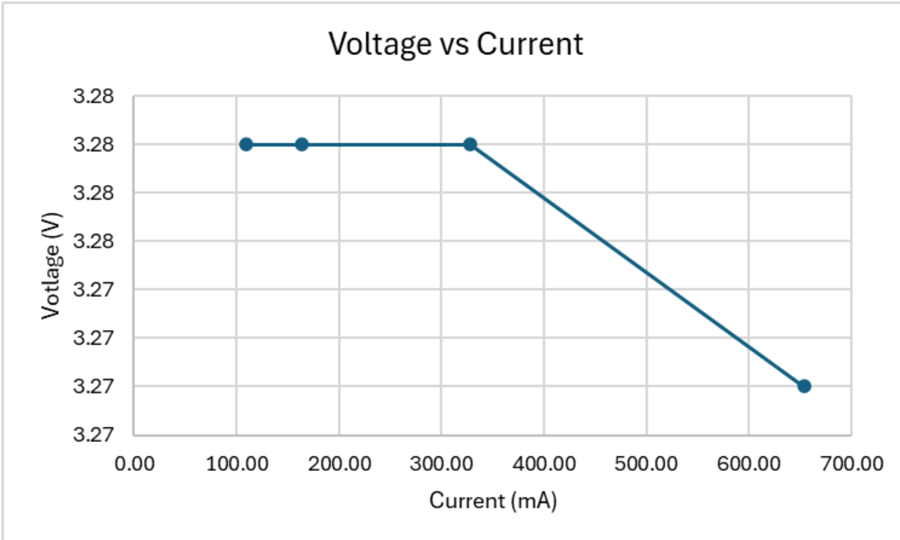
2.4.3. Prototype

The same process of building out the circuit on a breakout board and testing the current load took place with the LM2575-3.3WT. This ensured that as I designed this within the PCB, the circuit would be reliable and work as intended. Within Figure 3, the yellow box shows the 12 volt to 3.3 volt buck converter circuit. This was designed as compact as possible and it mirrored the 5 volt buck converter. This breakout board allowed for the design process to move on with confidence when designing the PCB.

2.4.4. Validation

Just as the 5 volt buck converter validation relied around current load testing, the 3.3 buck converter is the exact same. As the 3.3 volt buck converter is powering only the MCU and some of the multimeter and oscilloscope parts, and the signal generator, so the current expected output of the 3.3 buck converter was roughly 500 mA. As shown below in Figure 6, the current load testing went up to 650 mA just to make sure if any spikes came from the MCU, the buck converter could handle it.

Figure 6: Current Load Testing of 12 Volt to 3.3 Volt Buck Converter



In **Figure 6**, the 3.3 volt buck converter had very little voltage sag which was expected especially at low currents since the circuit is rated for 1 amp. The voltage dropped to 3.27 volts at the highest possible current needed. This gives a 0.9% error which more than meets the requirement needed for a safe, stable buck converter able to power everything needing the 3.3 volt supply.

2.5 5 Volt to -5 Volt Output

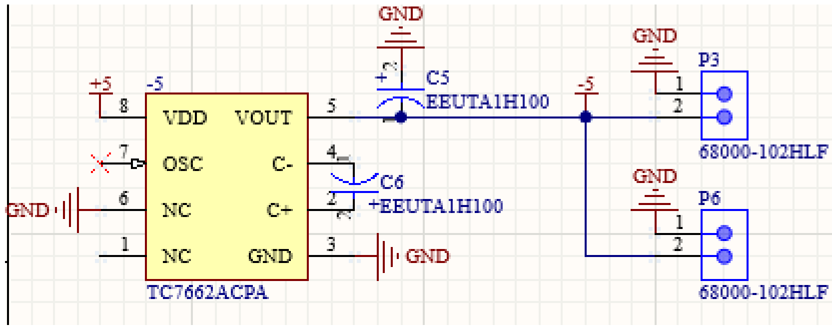
2.5.1. Operation

This subsystem is critical as well since the students will need a constant -5 voltage for their Op Amp circuits created within the further labs. The specific chip used with this system is the TC7662A, which allows for 5 volts to be inverted to -5 volts consistently. As with this circuit, the design schematic was taken from the data sheet and tested on the prototype board shown in Figure 3.

2.5.2. Schematic Design

The schematic design is quite simple as it only has two low cost capacitors within the circuit. This helps limit the need for an extra inductor and diode. As well, the circuit is very compact which helps save on cost of the entire design. A downside to this budget friendly design is that the output current is only rated for 40 mA, but this works well with our need since this -5 volt output is only used for Op Amps that require very little current.

Figure 7: TC7662A Schematic



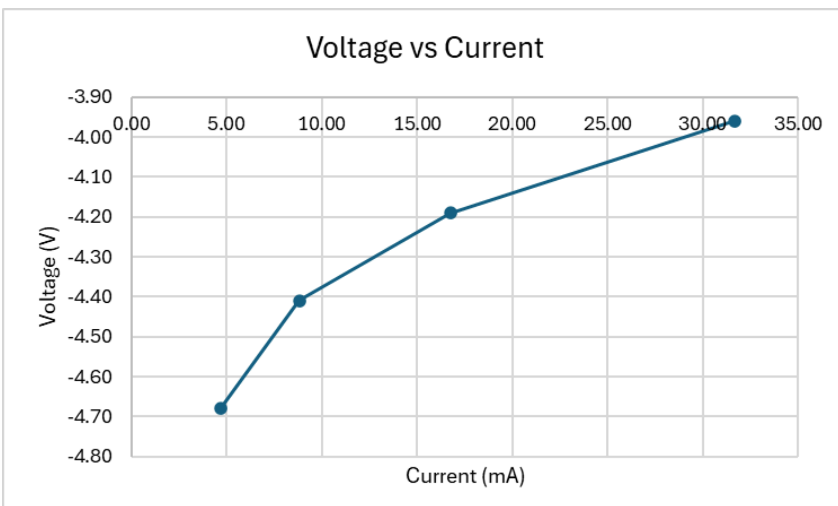
2.5.3. Prototype

As described in the 5 volt and 3.3 volt buck converters, in Figure 3 within the green box, the schematic was placed on a breakout board first since this allowed for me to test the system in person before placing it on a PCB. Under no load conditions on the breakout board, the circuit worked extremely well and was later used in the final design.

2.5.4. 5 Volt to -5 Volt Validation

This system needs to be constant and handle multiple different loads. To validate this system, it was tested from 5 mA to 32 mA and measured for the output voltage. This allowed us to understand how the chip would react under load and if it was sufficient enough for the required labs.

Figure 8: Current vs Output Voltage for the -5 Volt System



As shown in Figure 8, the -5 volt output drops down to -3.96 volts as the current gets closer to the max output current of 40 mA. This caused some issues within the measurement tools of the system since the multimeter and oscilloscope were expecting -5 volts, but to fix this issue we were able to change the scaling in the code. Overall, the -5 volt output was not exact, but was sufficient enough to not cause any clipping within any of the labs.

2.6 5 Volt Controlled Sweep

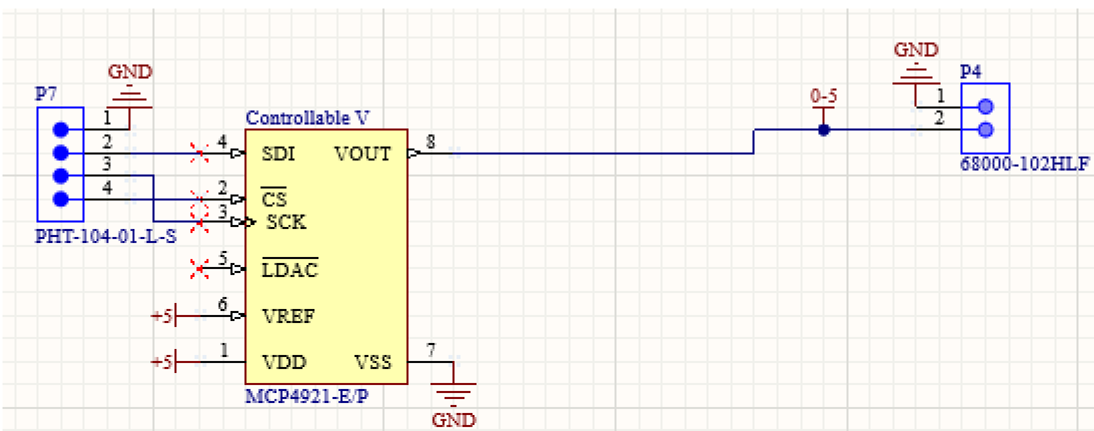
2.6.1. Operation

The students within almost all labs need to have different voltages ranging from 0 to 5 volts, and this needs to be constant for the students to have accurate measurements. This system, it uses a Digital to Analog Converter (DAC) with the chip MCP4921. This chip allows for the system to pick a voltage from 0 to 5 volts within a 50 mV accuracy. The output current for the DAC is rated for 50 mA. As well, this chip uses SPI connection to allow the MCU to control the output voltage.

2.6.2. Schematic Design

The reason this specific chip was chose is that the system uses only the chip to make the variable output voltage possible. The MCP4921 uses no inductors, capacitors, diodes, or resistors to have an accurate output voltage. As well, the SPI connection is much more reliable and faster.

Figure 9: MCP4921 Schematic Design



2.6.3. Prototype

With this system as well, I could solder it into the system of the breakout boards that was already built with the fixed voltages explained above. This allowed me to already be integrating the variable voltage with the 5 volt rail. As well, since the chip didn't need any other components, the only other setup was connecting the SPI connection to the ESP32. For the initial prototyping, I used a dev board kit, but later on we used a regular ESP32. This can be shown in Figure 3, within the blue box, where the entire power system is soldered into the breakout board.

2.6.4. DAC Code

The code for the variable voltage output was very minimal and easy to understand. Essentially, the student within the app would just input a voltage value and the DAC would set that voltage compared to the reference voltage which is 5 volts.

Figure 10: DAC Code

```
#include <SPI.h>

#define DAC_CS 5              // Chip Select pin

void setup() {
  Serial.begin(115200);
  SPI.begin();                  // SCK=18, MOSI=23 default for ESP32 VSPI
  pinMode(DAC_CS, OUTPUT);
  digitalWrite(DAC_CS, HIGH);

  Serial.println("Type a voltage (0..5) and press enter:");
}

void setDACVoltage(float voltage) {
  if (voltage < 0) voltage = 0;
  if (voltage > 5.0) voltage = 5.0;   // VREF=5.0V
  // MCP4921 is 12-bit (0..4095); At VREF=5.0V:
  uint16_t val = (uint16_t)(voltage * 4095.0 / 5.0 + 0.5); // Round to nearest
  uint16_t command = 0x3000 | (val & 0x0FFF);

  digitalWrite(DAC_CS, LOW);
  SPI.beginTransaction(SPISettings(20000000, MSBFIRST, SPI_MODE0));
  SPI.transfer16(command);
  SPI.endTransaction();
  digitalWrite(DAC_CS, HIGH);

  Serial.print("Output set to: ");
  Serial.print(voltage, 3);
  Serial.println(" V");
}

void loop() {
  static String inputString = "";
  while (Serial.available() > 0) {
    char inChar = Serial.read();
    if (inChar == '\n' || inChar == '\r') {
      if (inputString.length() > 0) {
        float voltage = inputString.toFloat();
        setDACVoltage(voltage);
        inputString = "";
      }
    } else {
      inputString += inChar;
    }
  }
}
```

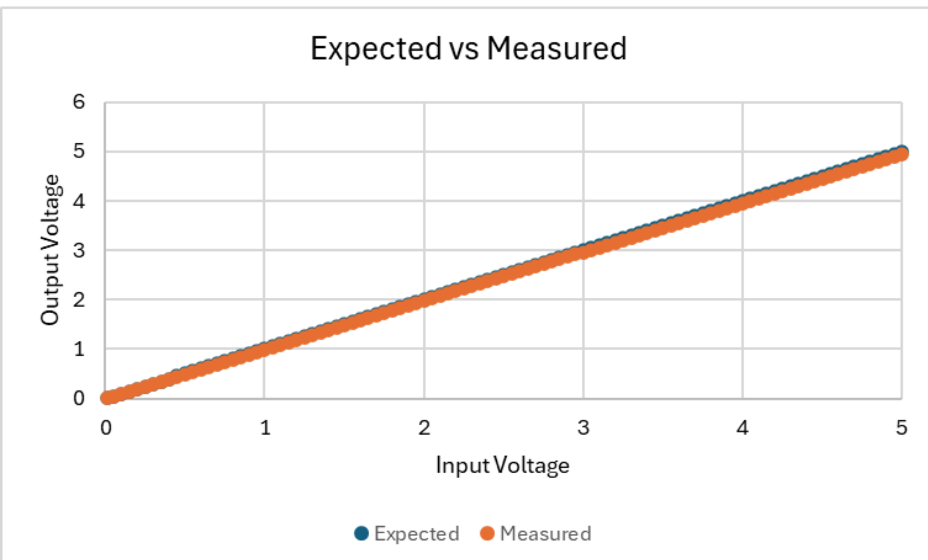
2.6.5. 5 Volt Controlled Sweep Validation

For the validation of this system, the current load testing wasn't enough to be just at 5 volt output. We tested the current load by choosing three different values and then increased the current as much as possible til we saw a voltage sag. As this voltage output is only rated for 40 mA, we weren't able to test to high values.

Table 1: Current Load Testing of Variable Voltage

Resistor (Ω)	Current (mA)	Input Voltage (V)	Output Voltage (V)	Power (mW)
10000	0.493	5	4.93	2.430
5000	0.982	5	4.912	4.826
3333	1.467	5	4.891	7.177
10000	0.326	3.3	3.261	1.063
5000	0.652	3.3	3.26	2.126
3333	0.977	3.3	3.257	3.183
10000	0.010	0.1	0.098	0.001
5000	0.020	0.1	0.098	0.002
3333	0.029	0.1	0.098	0.003

Figure 11: Accuracy of Digital to Analog Converter



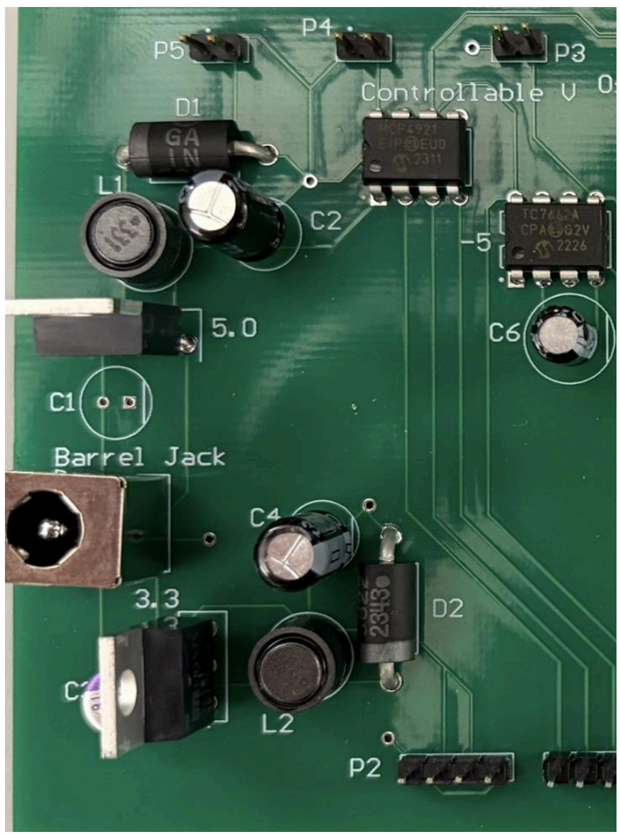
As shown in Table 1, the voltage sag is very minimal with the digital to analog converter. By switching over to the DAC instead of trying to use a digital potentiometer, the system is much more accurate and efficient. As well, the SPI connection works everytime to change the voltage not only from the computer, but also from the app. The accuracy ended up being around 99% where the end of the range near 4.5 voltages, there was some deviations. These were very minimal and would still allow the student to conduct the labs fully.

2.7 Subsystem Conclusion

2.7.1. PCB Design

After all of the prototyping and testing the entire power supply system was designed and built into a PCB where we were able to take measurements that are shown above. The process of combining all of the schematics onto one PCB was very efficient and for sake of time, we used JLCPCB which was also very cheap. The board was kept minimal to around 100 mm by 75 mm and it was only a 2-layer board. The board also allowed for the power to jumped to other boards and breadboards for testing and combining the system entirely.

Figure 12: Completed Power Supply PCB



2.7.2. Overall Functionality

The overall finished PCB board was very versatile as everything was functioning correctly just as the prototype was. With all of the extra jumper cables, the board could power multiple things at once for hours. As shown in Figure 11, the system was also very compact and took little space to operate. The system was very cheap and met all of the expect requirements from the DC Power Supply in the ECEN 215 Labkit.

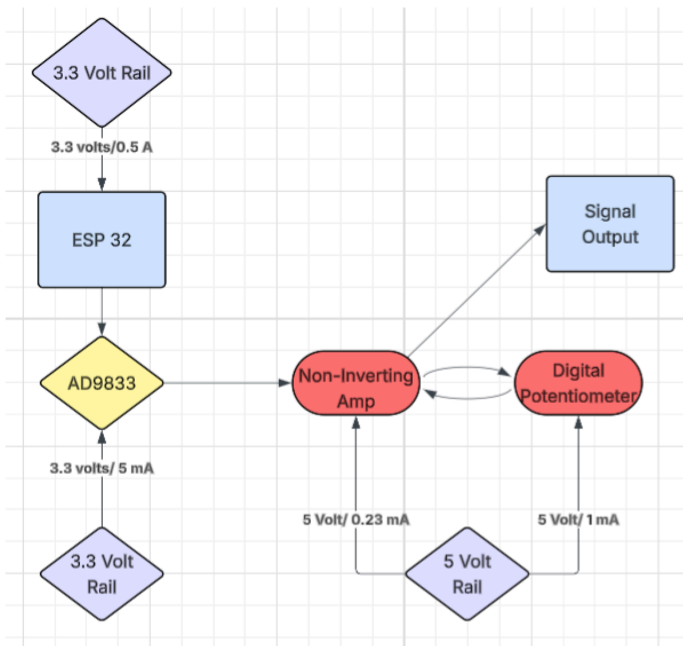
3. Signal Generator Subsystem Report

3.1 Subsystem Introduction

Within some of the labs, the students need to have a signal within their circuit. They need to be able to adjust the frequency of each signal from 20 Hz to 1,500 Hz. As well, they need to be able to change the voltage of the signal in a range of 0.5 to 2 volts. On top of that, they need to be able to change the waveform of the signal from sine, ramp, and square waves. This is a critical component of the system, so students can understand the aspects of a signal within circuits. For these reasons, this system needs to be accurate and validated for these three specs.

3.2 Subsystem Design

Figure 13: Subsystem Flow Chart



3.3 Controlled Voltage

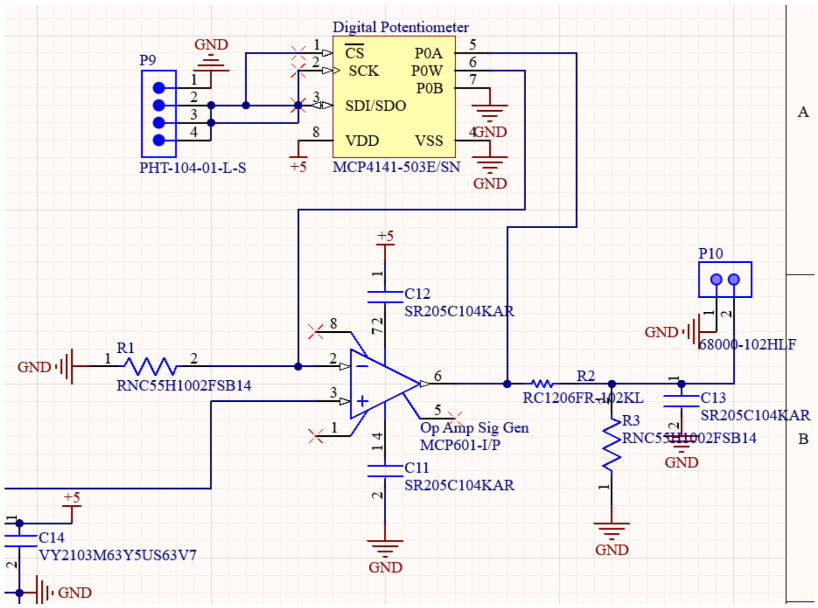
3.3.1. Operation

The controlled voltage of the signal is controlled by the digital potentiometer over the Op Amp circuit. The digital potentiometer is the MCP4161, which is powered by the 5-volt rail and switches the feedback resistance to control the gain. As the original output of the AD9833 is 0.6 volts, we placed a voltage divider to drop that to a clean 0.5 volts, which is the exact bottom of the range they need. With the potentiometer rated for 50,000 ohms, the circuit can exceed the initial 2-volt range.

3.3.2 Schematic Design

The design of the controlled voltage output centers around an Op Amp that has a digital potentiometer with an SPI connection. This allows the student to just change the feedback resistance and then change the gain to achieve the proper output peak-to-peak voltage.

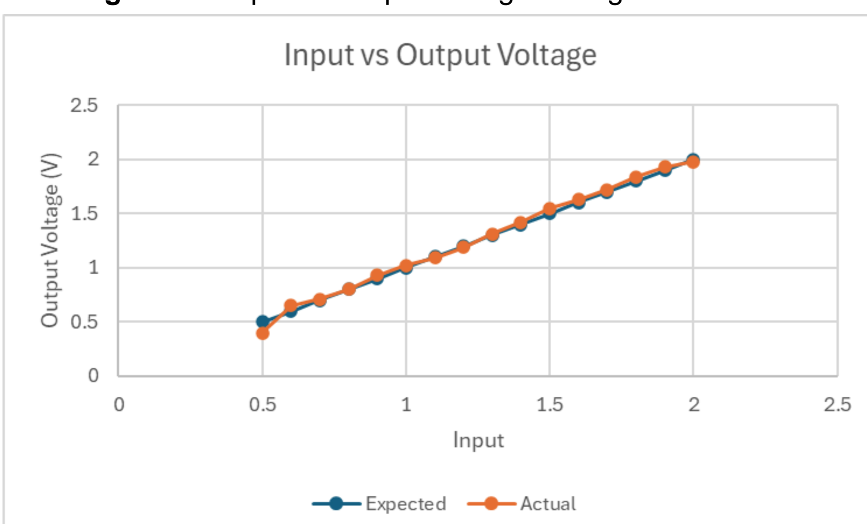
Figure 14: Controlled Voltage Schematic Design



3.3.3. Controlled Voltage Validation

For the validation of the controlled voltage, I essentially took an input voltage and made sure that I could amplify the voltage through the Op Amp. Keeping the Input voltage constant at 0.5 volts and then controlling the output voltage by setting the digital potentiometer, we can see in Figure 15 that this section of the Signal Generator is very accurate.

Figure 15: Input vs Output Voltage for Signal Generator



From Figure 15, the error of accuracy was roughly around 3%, which is a well-maintained margin. We can see that at lower and higher voltages, the system starts to deviate, which can be due to the voltage rails not being exactly 5 volts.

3.4 Frequency Range

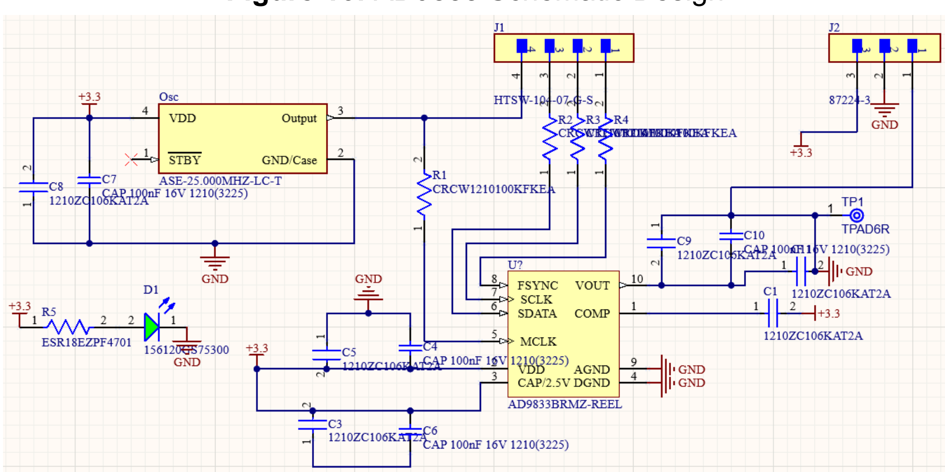
3.4.1. Operation

This system is responsible for creating the varied frequency output of the signal in the range of 20 Hz to 1,500 Hz. This is needed as most of the labs that involve signals have a specific frequency. The system uses the AD9833 signal generator chip and ASE-25.000MHZ-LC-T oscillator chip. As well, the AD9833 is connected to the ESP32 to be able to tune the frequency. Without this system, the signal generator would output only one frequency and not be able to operate for the specific circuits.

3.4.2. Schematic Design

The schematic shows an oscillator taking in 3.3 volts and supplying an oscillating output voltage with a 1.6 peak-to-peak voltage. As well, the frequency of the oscillator is 25 MHz, which the AD9833 can use to cover the entire 10 to 1,500 Hz. With the SPI connection, the student is able to change the frequency.

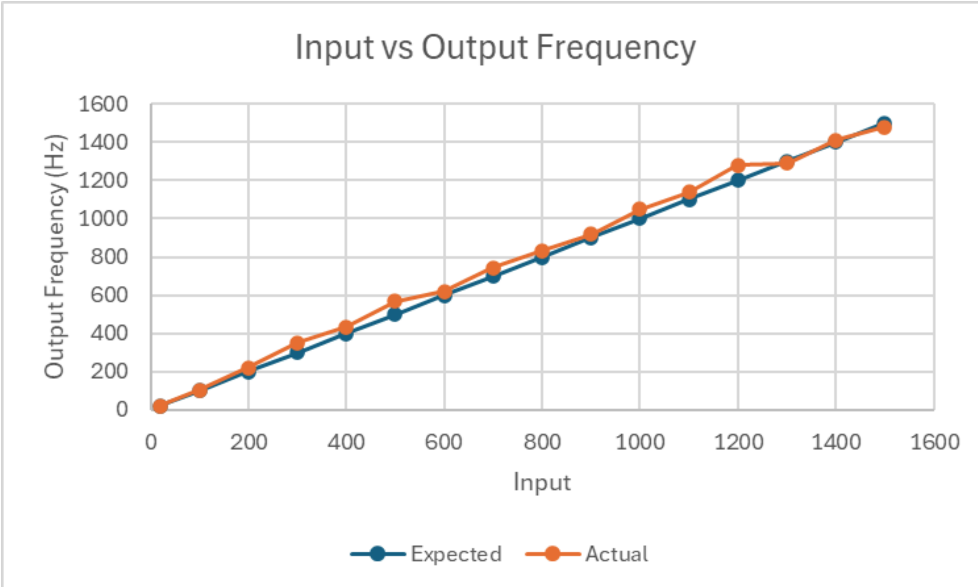
Figure 16: AD9833 Schematic Design



3.4.3. Frequency Range Validation

To validate this system by itself, it needed to be tested for each frequency output and how it correlated with the input from the ESP32. It was tested on a range of 20 Hz to 1,500 Hz, jumping 100 Hz every time.

Figure 17: Input vs Output Frequency for Signal Generator



The figure above shows that the expected output frequency is very close to the actual output frequency when the system is isolated. The system responded very well with thorough testing, and with each point in Figure 6, the actual frequency was accurate, and the expected frequency. For the future design, the noise coming from the oscillator to the AD9833 needs to be simplified to create a smoother connection for more accurate outputs.

3.5 Waveform Selection

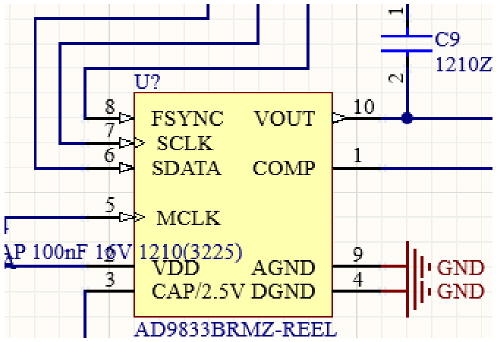
3.5.1. Operation

For the operation of this system, the student needs to be able to pick waveforms from Sine, Ramp, and Square, as most of the labs involving signals require these three waveforms. The AD9833 signal generator chip has the capability to output these specific waveforms. With the chip connected to the ESP32, the student will be able to select a needed waveform, which will then change the output of the AD9833.

3.5.2. Schematic Design

The design to change the waveform relies on the SPI connection to ESP32; the FSYNC pin shown in Figure 18 is the main pin that allows for waveform changes. Students within the code can pick the specific waveform they need for specific labs.

Figure 18: Schematic Design of Waveform Selection



3.5.3. Frequency Range Validation

The validation for this system is very simple, as it was tested to determine if it was able to output the specific signal needed and how well it was formed. If the student isn't able to change the waveform or the waveform is noisy, then the student won't be able to achieve accurate measurements.

Figure 19: Actual Output of a Sine Waveform

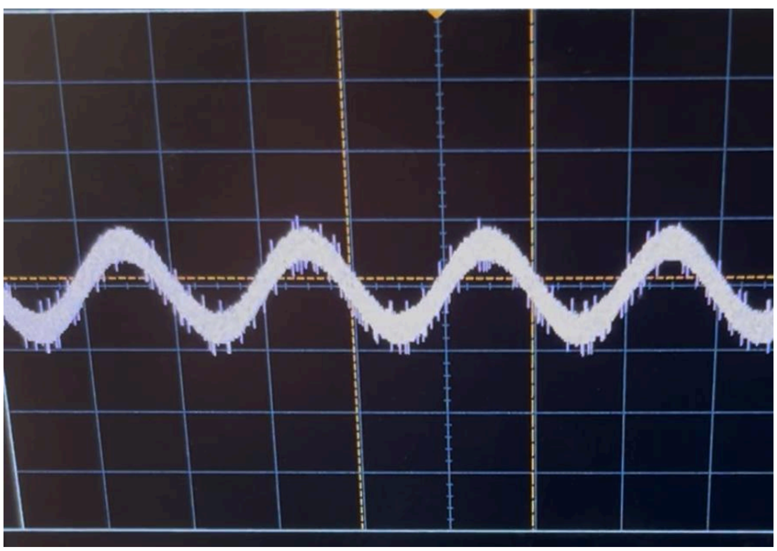


Table 2: Waveform Validation

Input	Expected	Output
Sine	Sine	Sine
Ramp	Ramp	Ramp
Square	Square	Square

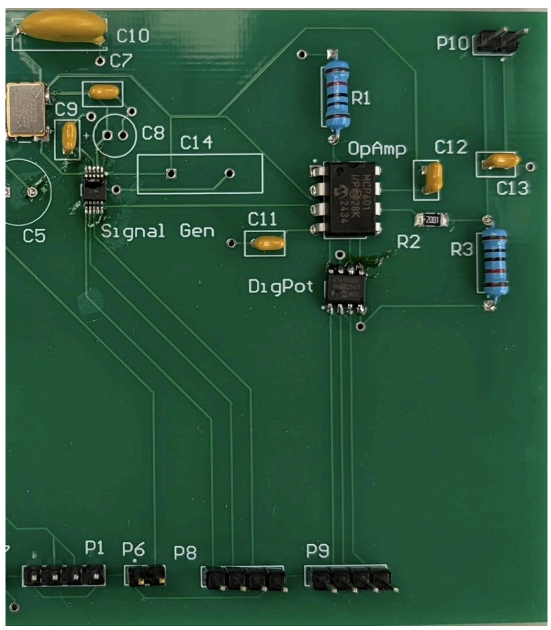
As shown in **Table 2**, the system was able to accurately output a specific waveform from either Sine, Ramp, or Square. By inputting from the ESP32, the AD9833 was able to change the output waveform. In Figure 19, the image is directly from one of the tests run for the system, and it shows that the waveform is accurate, but has some noise surrounding the Sine wave. With this noise, the students could have a hard time measuring certain aspects, so for future designs, this will need to be taken into consideration.

3.6 Subsystem Conclusion

3.6.1. PCB Design

By combining the 3.3-volt rail, the system is ready for integration and can operate as expected when working in isolation with an ESP32 dev board kit. The actual design of the ESP32 was larger than the power supply, as many capacitors and resistors were needed to filter out lots of noise in the system.

Figure 20: PCB Design



3.6.2. Signal Generator Code

The code I originally used to test the waveform switching and frequency changing was just a set loop where it changed with a delay between each iteration. This allows for validation needed within the isolated subsystem. The integrated code would allow a student to pick the needed waveform and frequency.

Figure 21: Signal Generator Code

```
#include <AD9833.h>
#define FSYNC   12    // Chip Select (FSYNC)
#define SCLK    18   // SPI Clock
#define SDATA   23   // SPI MOSI
AD9833 gen(FSYNC, SCLK, SDATA);

void setup() {
  Serial.begin(115200);
  gen.begin();
  delay(100);  // Added delay to stabilize
  gen.setFrequency(100);  // 100 Hz
  Serial.println("AD9833 Initialized");
}

void loop() {
  gen.setWave(AD9833_SINE);
  Serial.println("Sine wave");
  delay(10000);
  gen.setWave(AD9833_TRIANGLE);
  Serial.println("Triangle wave");
  delay(10000);
  gen.setWave(AD9833_SQUARE1);
  Serial.println("Square wave");
  delay(10000);
}
```

3.6.3. Overall Subsystem Conclusion

Overall, the system worked accurately and had similar outputs to the expected values. The controlled voltage worked well and had minimal noise for the system. The frequency response also worked well, but towards the end of the range, the noise increased more than expected. Finally, the waveform selection worked better than expected and allowed for different waveforms to be generated. The slight noise with each of these aspects is something that could be caused by bad soldering, or some values on the devices connected in these circuits might need to be adjusted.

4. Mobile App Subsystem Report

4.1 Subsystem Introduction

The mobile app subsystem is how the user interacts with the Labkit and completes labs. Since there are no buttons or interface devices on the Labkit, the app contains all functions relating to controlling the Labkit, as well as interfacing with the other subsystems, storing data, and exporting lab results.

4.2 Starting Screen

4.2.1. Starting Screen Details

The starting screen is what the user will see when first loading up the app on their device. The screen lets the user navigate to the connection screen, the tutorial screen, and the lab list screen.

4.2.2 Starting Screen Validation

Figure 22: Starting screen (Connected)

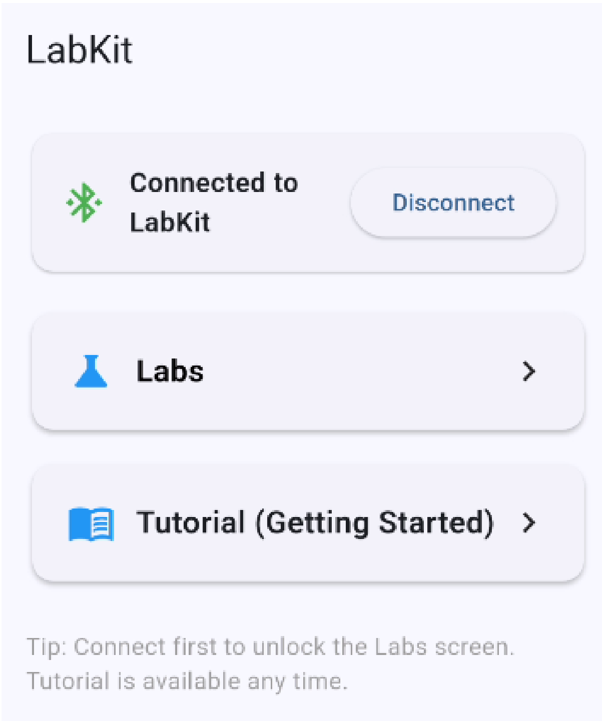
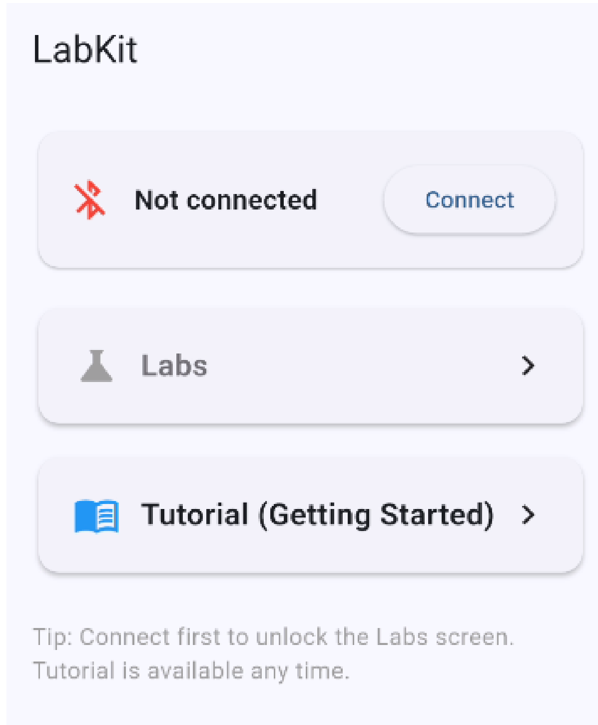


Figure 23: Starting Screen (Disconnected)



The Lab List screen can only be accessed once the app has connected to the Labkit. If the user wants to read the labs without the device, they will use the lab manual provided by the class.

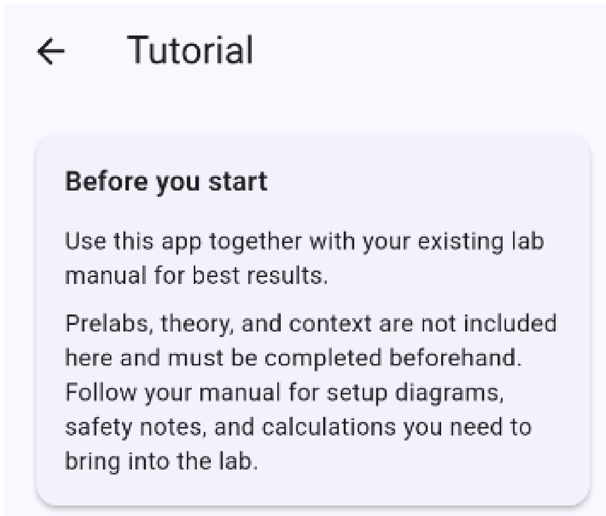
4.3 Tutorial Screen

4.3.1. Tutorial Screen Details

The tutorial screen explains how to use the multimeter and what format the oscilloscope will be in when doing the labs. It also explains the scope of what is contained in the app and what parts will need to be completed using the lab manual pdf.

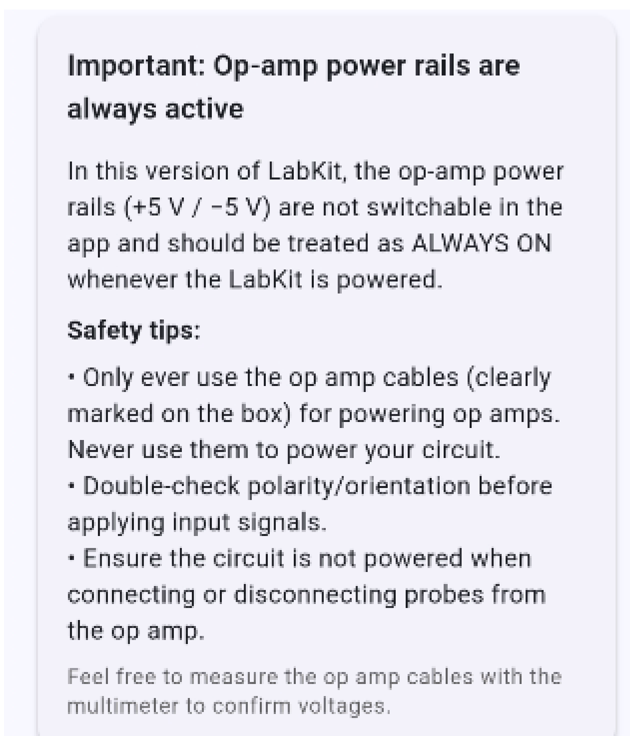
4.3.2 Tutorial Screen Validation

Figure 24: Before You Start Guide



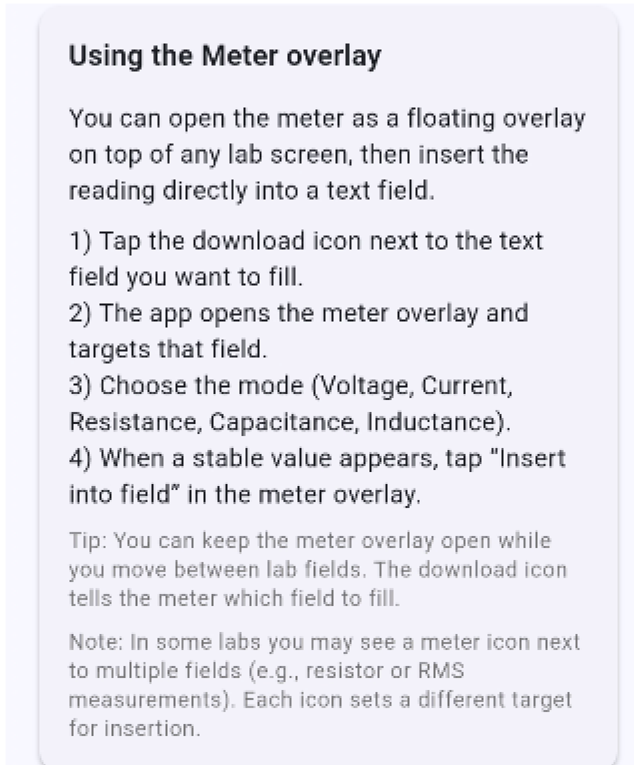
This section explains that the lab manual is needed to complete the labs completely and where to find helpful information that may be needed for certain parts of the labs.

Figure 24.1: Safety Warning



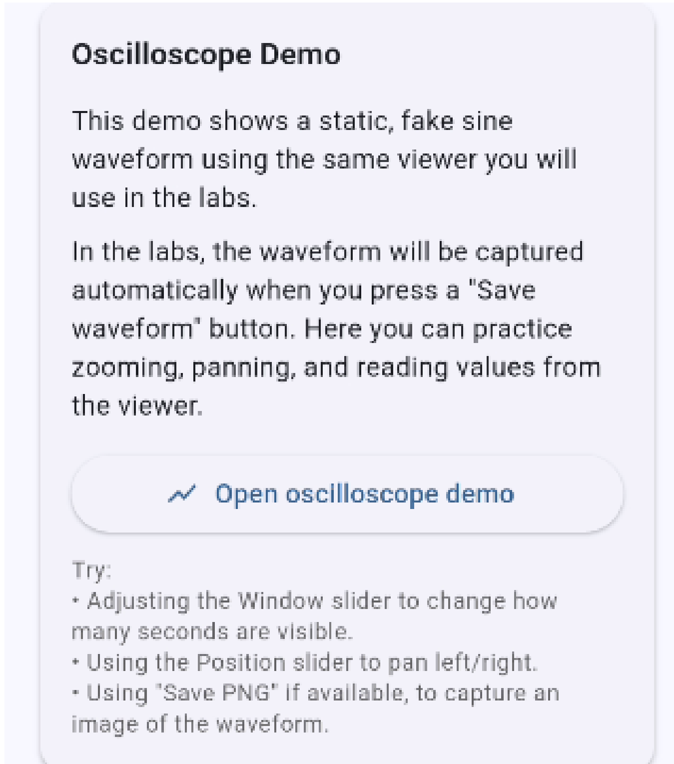
This section explains that the cables meant to power any op amps that may be required during the lab will stay set to a constant +5 and -5. This cannot be changed using the Labkit or the App so the user must be careful and only use the op amp cables for their intended purpose.

Figure 25: Multimeter guide



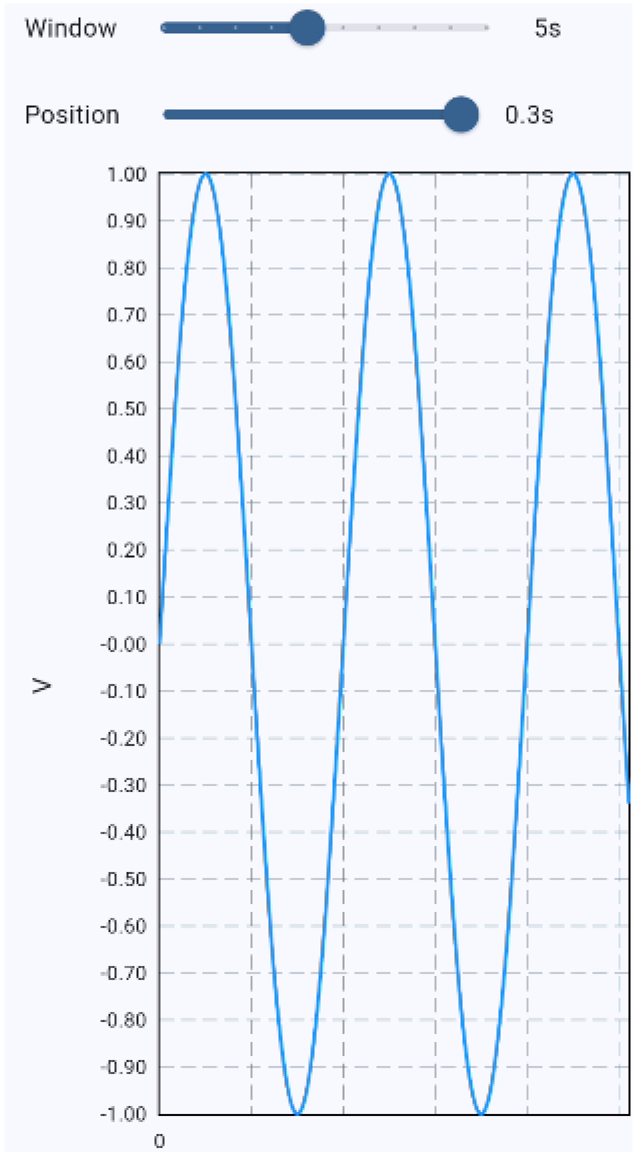
This section explains how to use the multimeter during the labs. This will be further explained in the multimeter validation later in this report

Figure 26: Oscilloscope Guide



This section explains how to correctly record and view waveforms in the lab. The open oscilloscope demo button provides a sample waveform similar to one that might be encountered in the labs (figure 14.1).

Figure 26.1. Oscilloscope Demo



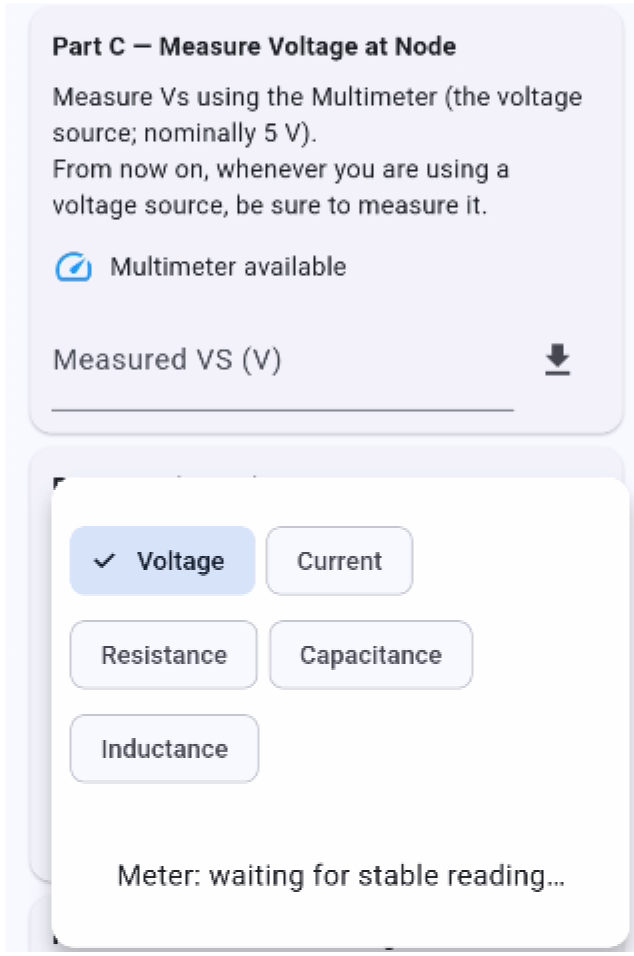
4.4 Multimeter

4.4.1. Multimeter Details

As previously mentioned in the tutorial screen, the multimeter allows the user to save data recorded from the circuits that will be built in the labs. It will open at the bottom of the screen and allow the user to navigate through the labs while also measuring various things.

4.4.2 Multimeter Validation

Figure 27. Multimeter Overlay



The screen in the background can be scrolled freely while the user connects the multimeter probes to their circuit. The type of meter can be freely changed using the buttons and the number can be easily inserted into any text fields in the labs so the user doesn't have to copy over long decimals.

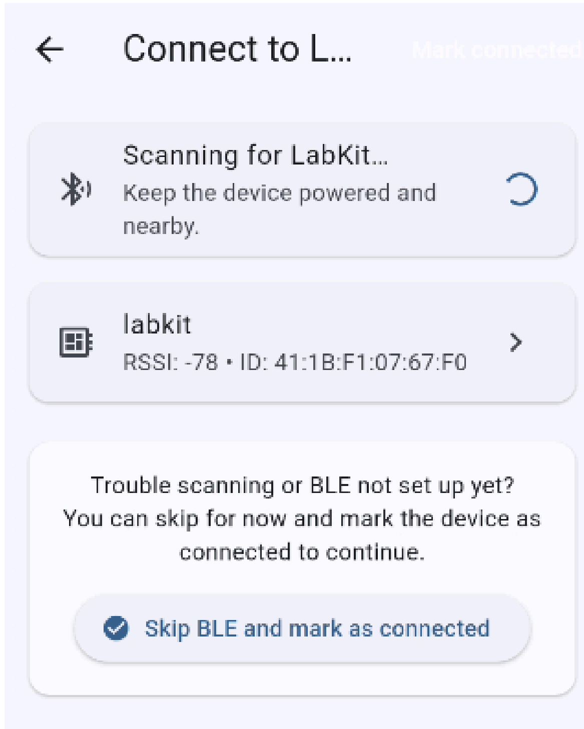
4.5 Connection Screen

4.5.1. Connection Screen Details

The connection screen is how the user connects the app with their Labkit. The built-in bluetooth antenna in the user's phone will scan for the BLE signal emitted from the Labkit and connect to the device.

4.5.2 Connection Screen Validation

Figure 28. Connection Screen



The app automatically begins scanning when the user presses the connect button from the starting screen. Once the Labkit's BLE signal is detected, it will show up in the list as 'Labkit'. To connect, simply select the Labkit. To skip connecting to the Labkit and instead continue on to the labs, the skip BLE and mark as connected button can be used though any features that require the Labkit (oscilloscope, multimeter, etc) will not be functional.

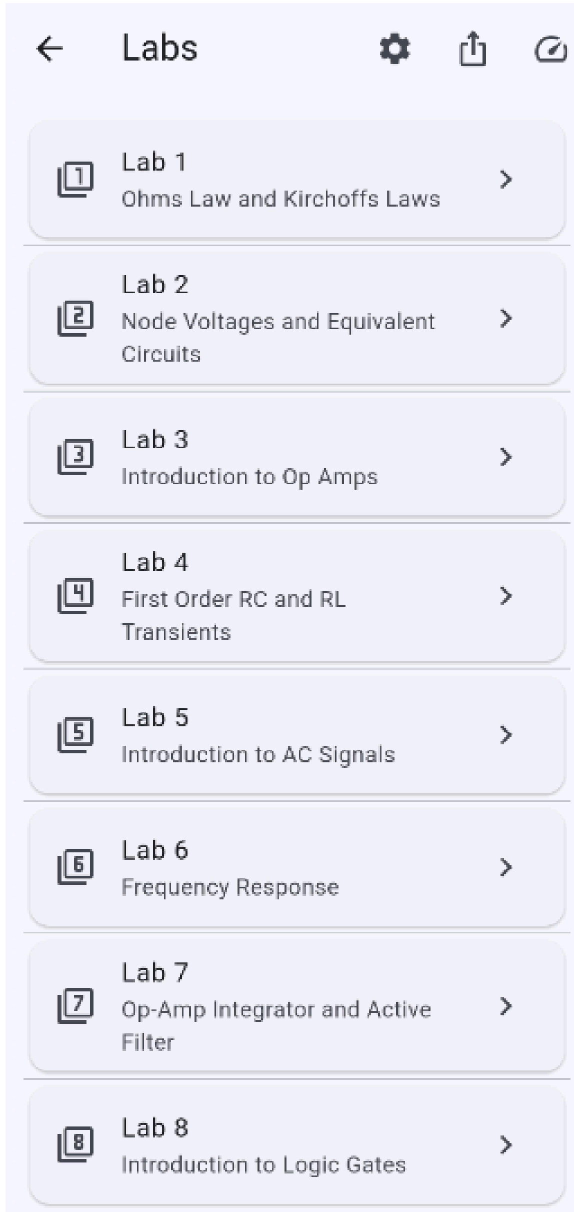
4.6 Lab List Screen

4.6.1. Lab List Screen Details

The lab list screen has multiple uses. The main use is to select which lab they wish to complete. This screen is also how the user accesses the settings screen, the export screen, as well as a button to open the multimeter for use outside of labs.

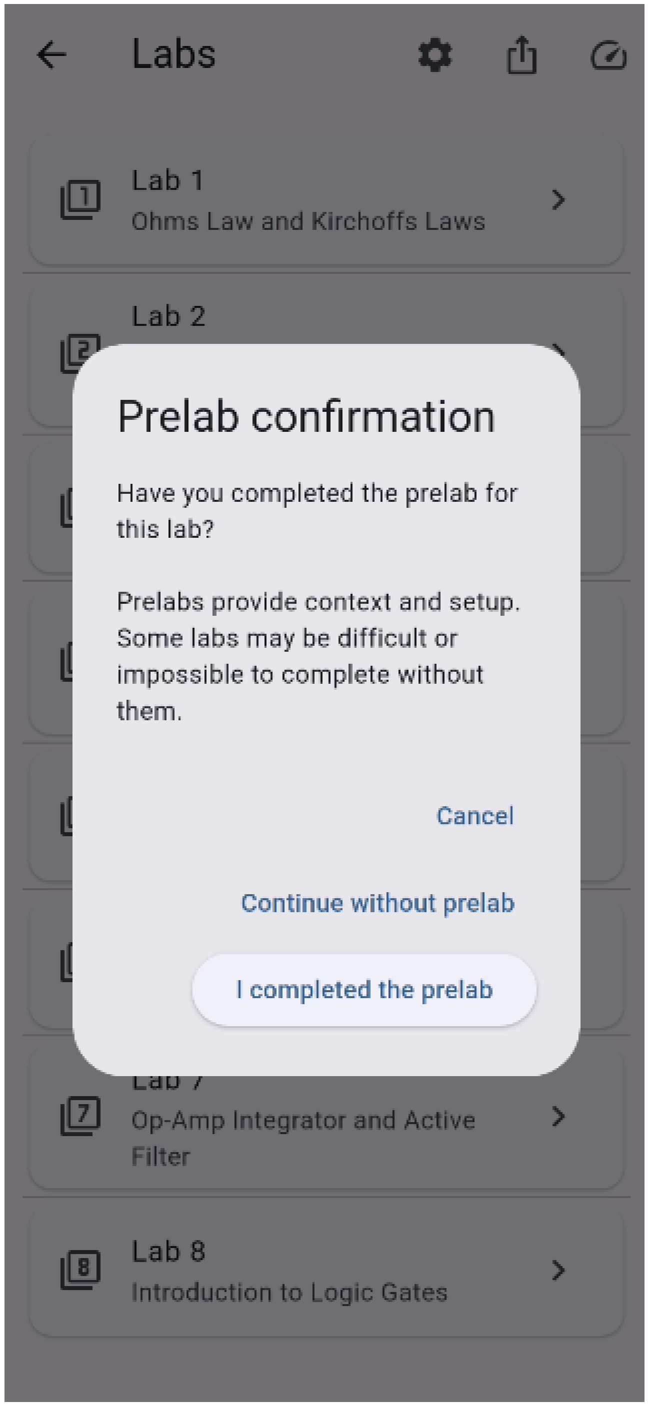
4.6.2 Lab List Screen Validation

Figure 29: Lab List Screen



The buttons at the top are (in order), the settings screen, the export screen, and the multimeter overlay toggle. These screens will be covered in further detail later in this report.

Figure 30: Prelab Check



When the user selects a lab, a confirmation message will appear to remind the user to complete the prelab before starting the main lab. The screen can be bypassed if needed by selecting 'continue without prelab'.

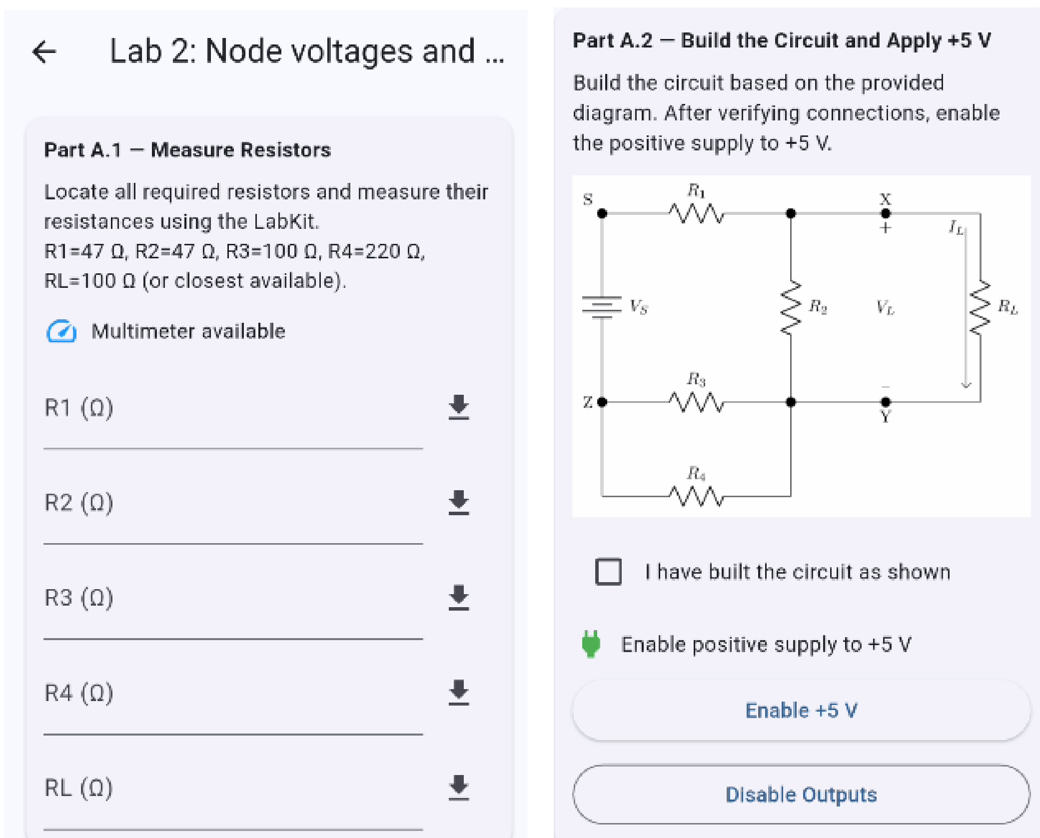
4.7 Individual Lab Screens

4.7.1. Individual Lab Screens Details

Each lab (1-8) is rewritten from the lab manual pdf to give more accurate and simpler instructions. In the pdf, the AD2 is used but has the capability to do much more than the scope of ECEN 215. This can be confusing so all labs were rewritten to be easily understandable and simple to complete with the Labkit. The app also provides tables and text fields so the user can be sure they have completed all parts of the lab.

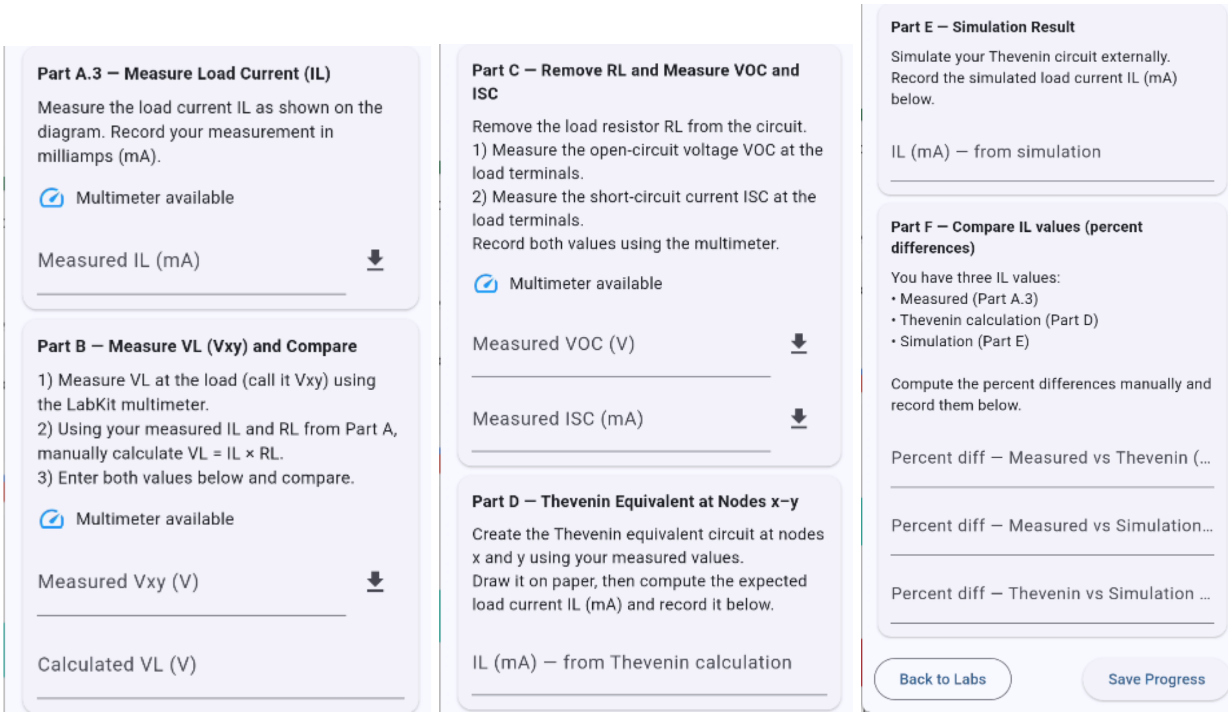
4.7.2 Individual Lab Screens Validation

Figure 31.1: Lab 2-1



In many labs the user will first measure all components needed to build the circuit to ensure all components are functioning correctly. In this case there are 5 resistors to measure. The user will then build the circuit(s) based off of the diagram(s). Once the circuit is built, the user enables the power supply. This ensures that the circuit can only receive the correct voltage to limit danger and reduce the amount of ways the components could be damaged.

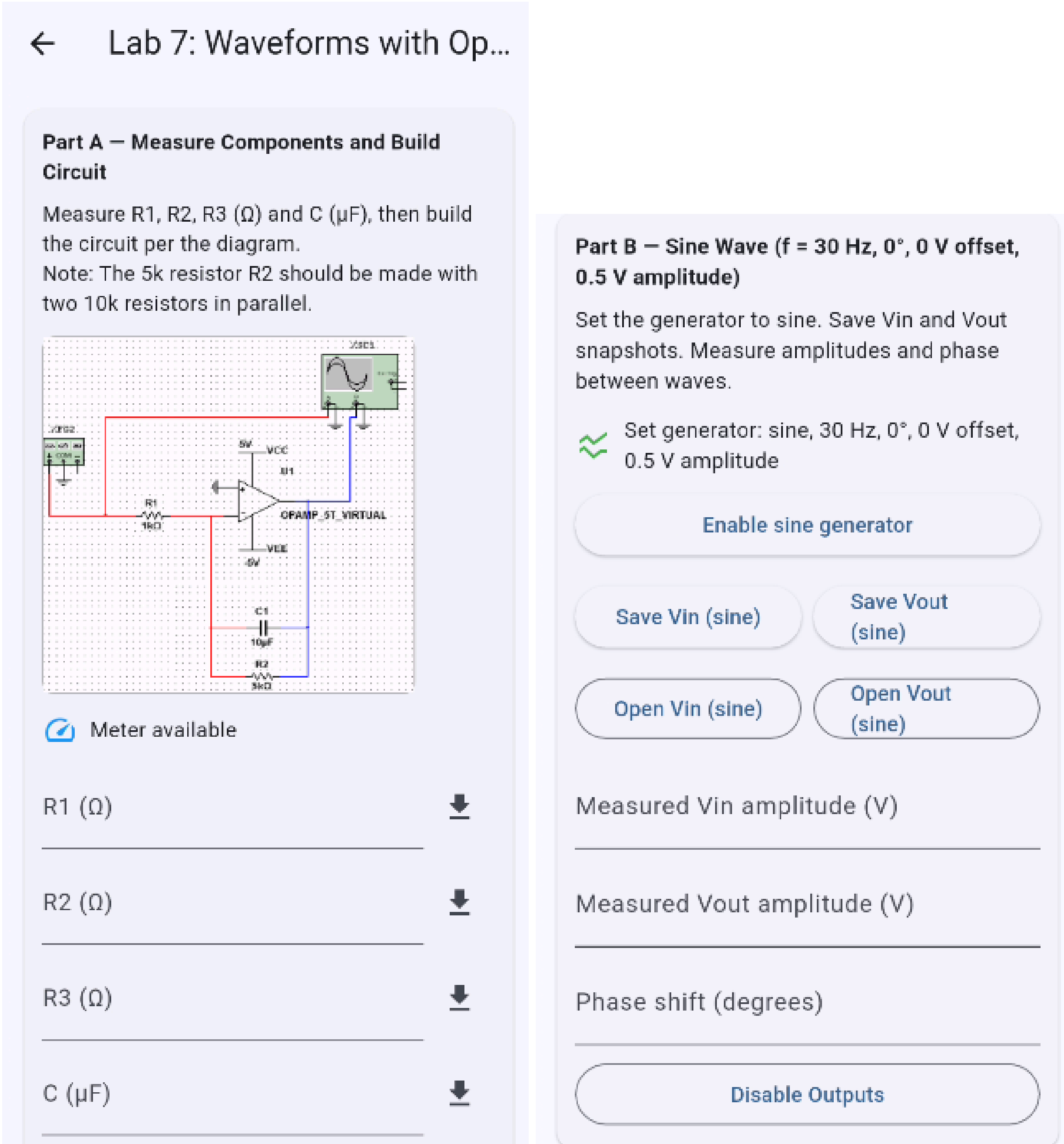
Figure 31.2: Lab 2-2



Once the circuit is complete, the user will answer various questions and record values from various parts of the circuit using the multimeter. Once all sections have been completed, the user will save their progress with the button on the bottom and can return to the lab list screen and complete other labs.

In this example, lab 2 is used to demonstrate what the labs generally look like. All parts of labs 1-8 are included in the app.

Figure 32: Lab 7



Some labs, such as lab 7, contain op amps which require both positive and negative voltage. The user will apply the dedicated op amp cables to the positive and negative as per the pin diagram of the specific op amp.

Lab 7 also contains parts with the oscilloscope and signal generator. The user will not customize the settings of the signal generator. Instead, the specifics of the signal, frequency, amplitude, offset, etc, will be pre-selected according to the lab. The user will enable the signal and choose to save the waveforms once the probes have been placed in the correct locations.

4.8 Settings Screen

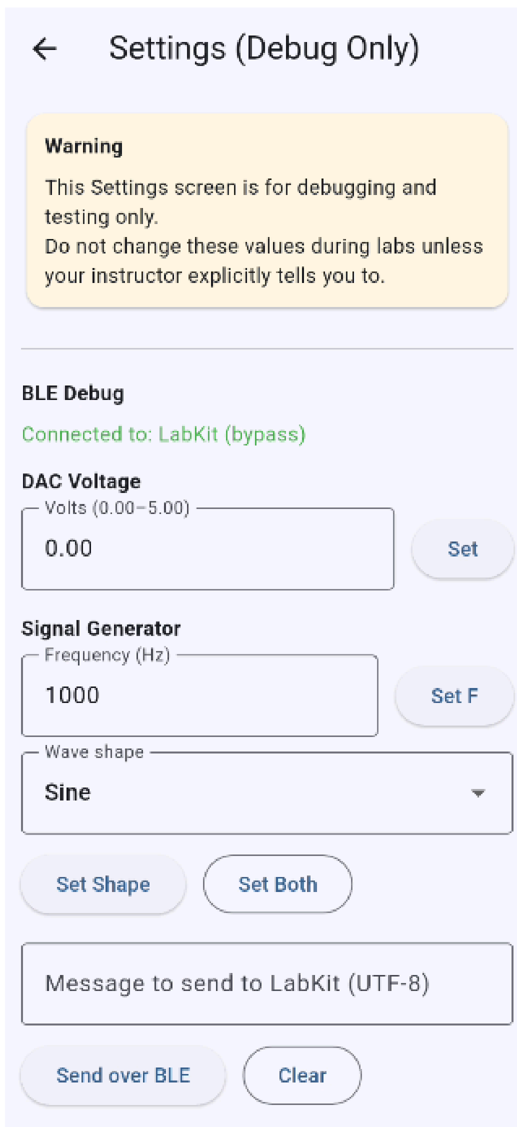
4.8.1. Settings Screen Details

The settings screen is where the user will customize anything that may be needed during the lab process. Currently, the only setting available is setting the email the user will use for exporting the data.

The settings screen is only to be used for testing and debugging purposes. When opening the settings screen a warning will be displayed to ensure the user knows that this screen is not required for any labs.

4.8.2 Settings Screen Validation

Figure 33: Settings Screen



These debug settings can be used to set specific outputs that may not be required for any specific lab. The output voltage can be set anywhere between 0-5 V. The frequency can be set between 0-10k Hz. The wave shape can be changed between sine, square, and triangular. The message to Labkit section can send specific commands or messages to the Labkit to confirm the connection between the app and Labkit, though the message cannot be read without the Labkit being connected to a computer.

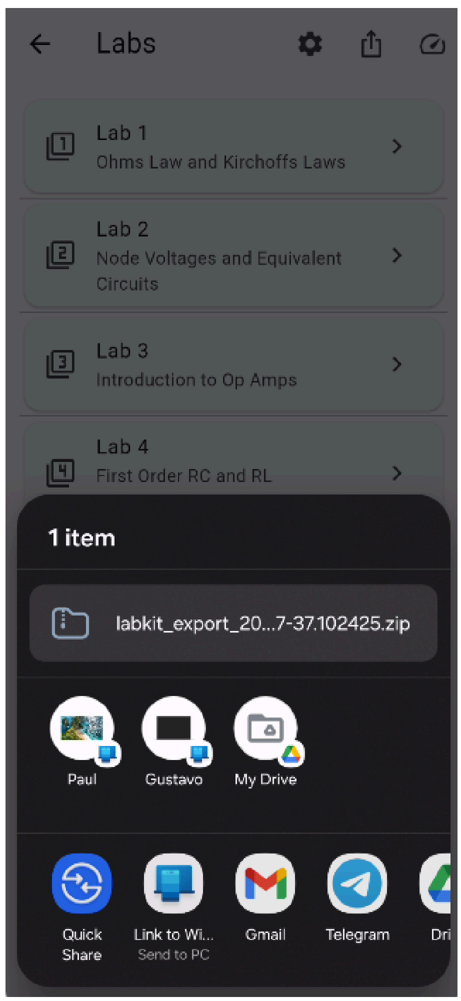
4.9 Export Screen

4.9.1. Export Screen Details

When the export button (at the top of the lab list screen) is pressed, an overlay will appear similarly to the multimeter overlay. The user is able to export through any app they may have on their phones if they don't want to send the file through email.

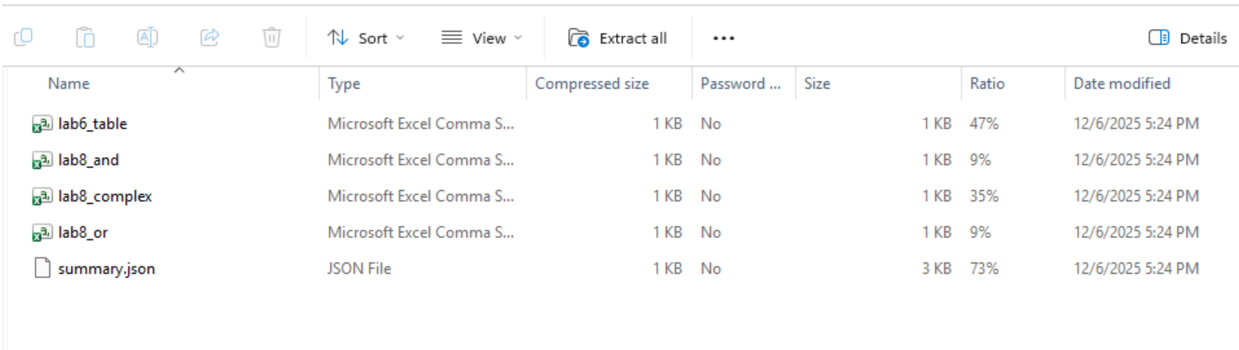
4.9.2 Export Screen Validation

Figure 34: Export Screen



The overlay itself goes through the user's individual device. Depending on what apps the user has on their phone, the export screen will look different.

Figure 35: Export File



Name	Type	Compressed size	Password ...	Size	Ratio	Date modified
lab6_table	Microsoft Excel Comma S...	1 KB	No	1 KB	47%	12/6/2025 5:24 PM
lab8_and	Microsoft Excel Comma S...	1 KB	No	1 KB	9%	12/6/2025 5:24 PM
lab8_complex	Microsoft Excel Comma S...	1 KB	No	1 KB	35%	12/6/2025 5:24 PM
lab8_or	Microsoft Excel Comma S...	1 KB	No	1 KB	9%	12/6/2025 5:24 PM
summary.json	JSON File	1 KB	No	3 KB	73%	12/6/2025 5:24 PM

The file the user receives will look something like this, depending on which labs were completed. It will contain excel files for each of the waveforms saved during the labs as well as a summary containing all the saved values and typed responses.

Figure 36: Export Summary


```
"generated_at": "2025-11-28T12:33:13.912564",
"student_email": "ecslyter@gmail.com",
"labs": {
 "lab1": {
  "rA1Ohm": 56.0,
  "rA2Ohm": 100.0,
  "circuitBuilt": false,
  "vCVolt": 5.0,
  "vE1VoltCalc": 4.6,
  "vE1VoltMeasure": 4.45,
  "vE2VoltCalc": 5.2,
  "vE2VoltMeasure": 5.0,
  "notesD": null,
  "notesF": "12"
 },
 "lab2": {
  "r1Ohm": 50.0,
  "r2Ohm": 100.0,
  "r3Ohm": 300.0,
  "r4Ohm": 1000.0,
  "rLOhm": 675.0,
  "circuitBuilt": false,
  "iL_mA": 20.0,
  "vxyVolt": 4.89,
  "v1CalcVolt": 5.0,
  "vocVolt": 5.3,
  "isc_mA": 5.0,
  "ilTh_mA": 20.0,
  "ilSim_mA": 20.0,
```

This export file contains a record of everything the user has done in all of the labs. Anything labeled 'null' means the user did not interact with that part of the lab. All data is stored locally on the user's device before being exported. This is meant to be combined with the waveforms as well as the prelabs the user completed outside of the app in a final lab report to be submitted by the user. This lab report is not included in the app.

4.10 Subsystem Conclusion

The mobile application subsystem successfully supports Android devices and contains multiple features to assist the user with completing the labs 1-8 in the ECEN215 class. This app is meant to be as simple as possible so the user is only required to learn topics in the scope of the 215 class. The app connects to the main Labkit system through Bluetooth Low Energy (BLE). This app is to be used alongside the lab manual pdf provided through the class as well as any supplementary material that may be needed per lab.

5. Multimeter Subsystem Report

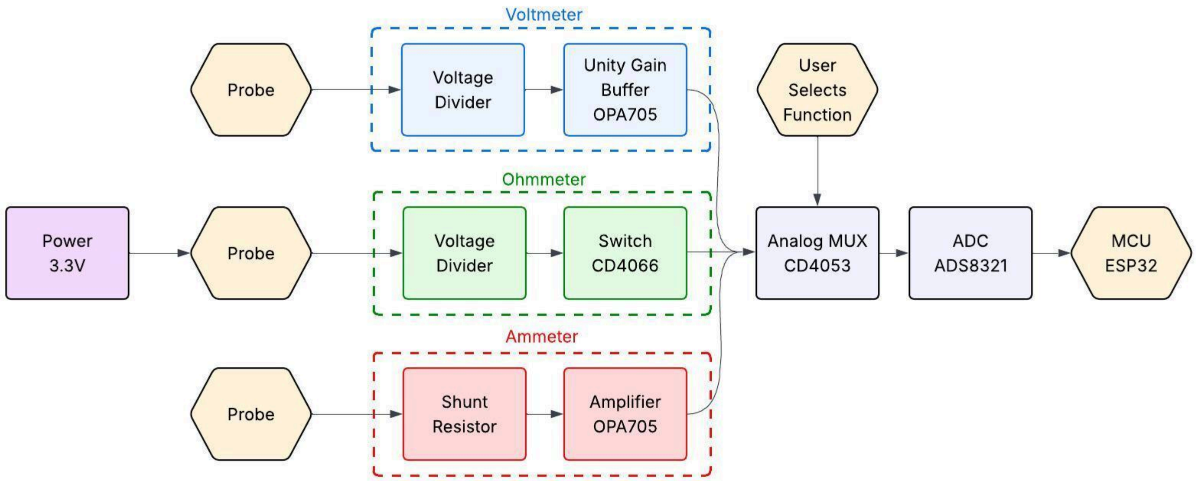
5.1 Subsystem Introduction

The multimeter subsystem is designed to measure voltage, resistance, and current of the user's circuits. Designed and implemented from the schematic level through final PCB fabrication, the multimeter ensures accurate and repeatable measurements across different ranges, which are sent to the kit's ESP32 microcontroller in order to communicate with the Mobile App subsystem. This subsystem is powered by the power supply subsystem, using common power rails on the PCB.

5.2 Subsystem Details

5.2.1 Subsystem Block Diagram

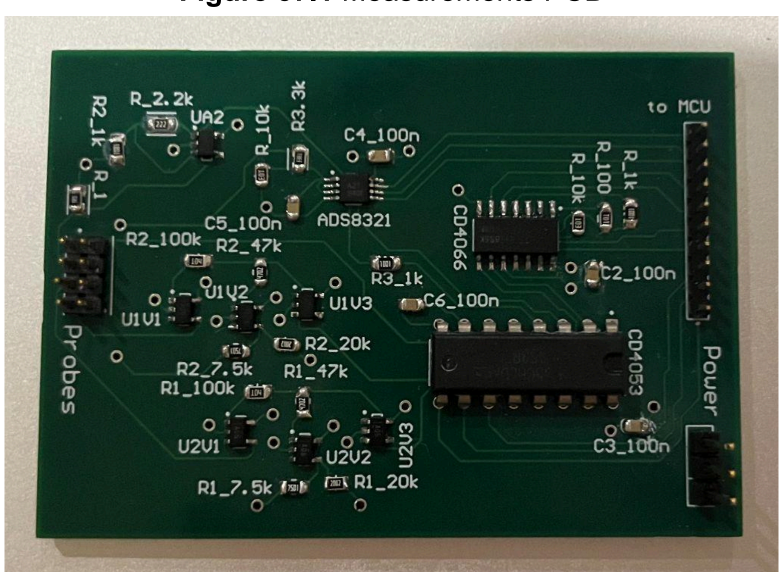
Figure 37. Multimeter Subsystem Block Diagram



As seen in **Figure 25**, the block diagram above, the multimeter subsystem uses input probes for each multimeter function, which pass through their individual paths and to the CD4053 MUX. Through the mobile app, the user selects their desired function, which allows the chosen function to be passed through the ADC, MCU, and back to the user through the mobile app.

5.2.2 Assembled PCB

Figure 37.1 Measurements PCB



5.2.3 Subsystem Schematic Design

Figure 38. Annotated Multimeter Subsystem Schematic

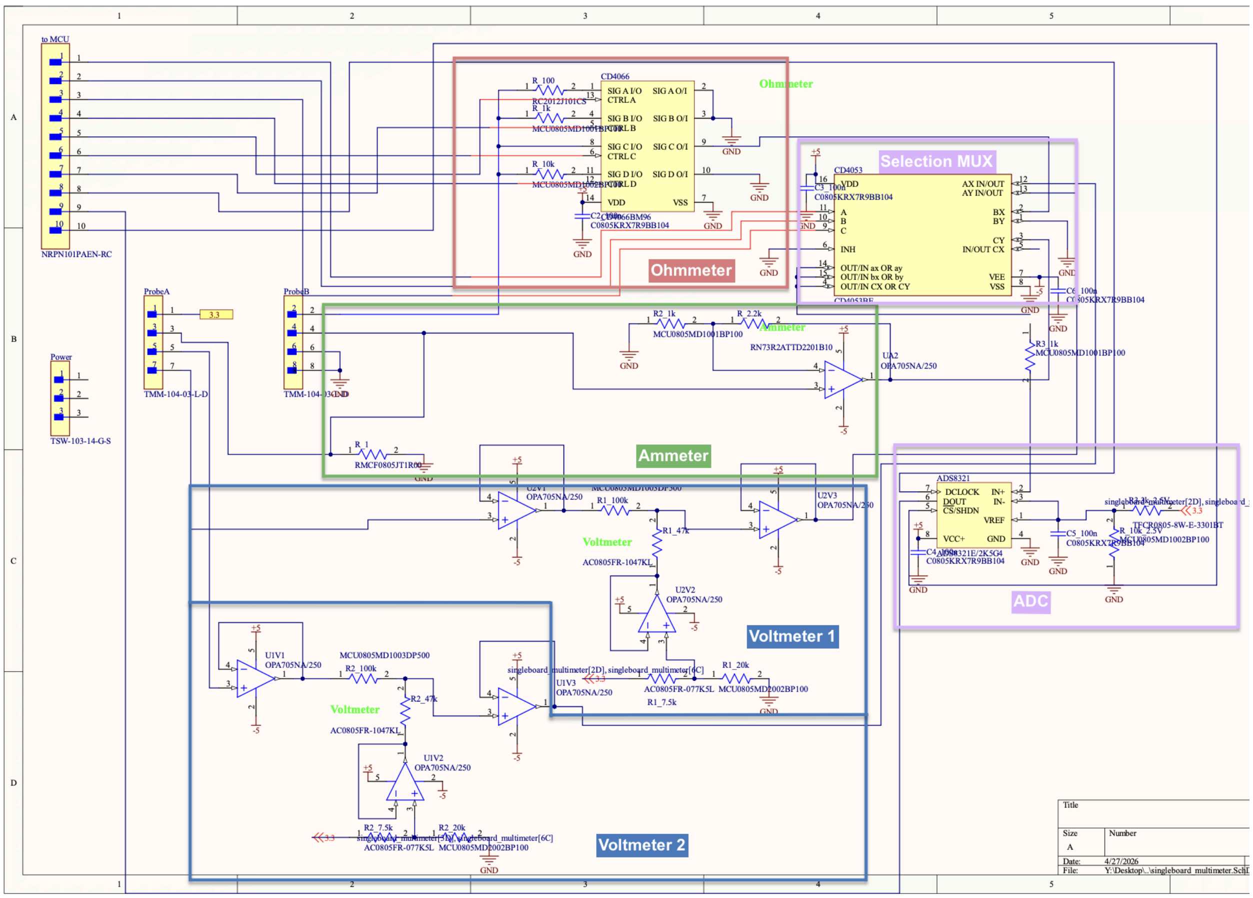


Figure 26 provides a subsystem overview of the multimeter designed to support voltage, current, and resistance measurements. Input signals first pass through scaling networks, such as voltage dividers and shunt resistors, to ensure safe operation across the intended measurement ranges. For current measurements, a shunt resistor converts current to a measurable voltage, while resistance measurements are performed by applying a known voltage and measuring the resulting voltage across a voltage divider. These signals are then passed through amplifiers and filtering stages to improve accuracy and reduce noise before being read by the system's analog-to-digital converter.

5.3 Voltmeter

5.3.1 Operation

The voltmeter is designed to measure between -5 and 5 Volts, using a voltage divider with a 2.4V offset to scale the voltage down to a MCU-safe range, which is 0 to 3.3V.

5.3.2 Validation

The voltmeter was tested at different voltages between -5 to 5 Volts in order to observe the output voltage, with 15 data points averaged at each voltage.

Table 3. Voltmeter Validation

V INPUT	V MEASURED	% ERROR
[V]	[V]	
-5	-5.086	1.7200
-4	-4.029	0.7250
-3	-3.058	1.9333
-2	-1.983	0.8500
-1	-1.019	1.9000
0	0.135	0.1350
1	1.022	2.2000
2	2.039	1.9500
3	3.068	2.2667
4	3.928	1.8000
5	4.964	0.7200

As seen in **Table 3**, across the 10V range, the expected and measured outputs were roughly similar, with the outputs within 2.5% of the expected values.

5.4 Ohmmeter

5.4.1 Operation

The ohmmeter is designed to measure between 10Ω and 10kΩ, using three ranges of voltage dividers to accommodate the varying current at different resistances. By first sending a known 3.3V into the circuit, then measuring between the unknown and known resistor, the MCU decodes the output values to determine the unknown resistor value using a rearranged voltage divider equation.

5.4.2 Validation

The ohmmeter was tested at different voltages within each resistance range in order to observe the output voltage. The probes of the ohmmeter were placed at each end of the resistor under test, and the results for each resistor were averaged 15 times and compared to the bench ohmmeter. In order to validate the software aspect of the ohmmeter, resistances at the cap of each range were tested. By iterating the measurements multiple times, the software was adjusted to better accommodate the edge cases.

Table 4. Ohmmeter Validation

Ω INPUT	Ω MEASURED	% ERROR
[Ω]	[Ω]	
10	10.193	1.9300
30	31.387	4.6233
50	52.328	4.6560
100	98.912	1.0880
300	304.294	1.4313
500	494.052	1.1896
1000	1018.577	1.8577
3000	3061.150	2.0383
5000	4903.409	1.9318
10000	9824.336	1.7566

As seen in **Table 4**, across the range, the resistances after mathematical adjustment were within 5% of the expected value.

5.5 Ammeter

5.5.1 Operation

The ammeter is designed to measure between 0 and 1A, using a shunt resistor and gain amplifier to convert the current to voltage.

5.5.2 Validation

The ammeter was tested at different currents between 0 and 1 Amps in order to observe the output voltage. The ammeter was placed in series with a test circuit of varied resistance in order to test at different currents, then verified against the bench ammeter. The results were averaged 15 times for each current value.

Table 5. Ammeter Validation

I INPUT	I MEASURED	% ERROR
[A]	[A]	
0.001	0.0011	10.0000
0.01	0.0089	11.0000
0.1	0.105	5.0000
0.2	0.214	7.0000
0.4	0.374	6.5000
0.5	0.518	3.6000
0.6	0.638	6.3333
0.8	0.886	10.7500
1	1.362	36.2000

As seen in **Table 5**, across the current range, the expected and measured outputs were roughly similar, with most errors around 10%, but an edge case slightly higher but within 0.3A of the expected value.

5.6 Subsystem Conclusion

5.6.1 404 Changes

Between 403 and 404, the multimeter designs were adjusted to better align with the affordability of the device. By reducing the number of unity gain buffers, we trade off voltage changes in the millivolts for ~20% cost decrease of the subsystem.

5.6.2 Conclusion

Each component of the multimeter, the voltmeter, ohmmeter, and ammeter performed roughly as expected. Though measured and expected outputs had varying levels of error, each of the measurement tools show accuracy high enough for the lab course's purpose.

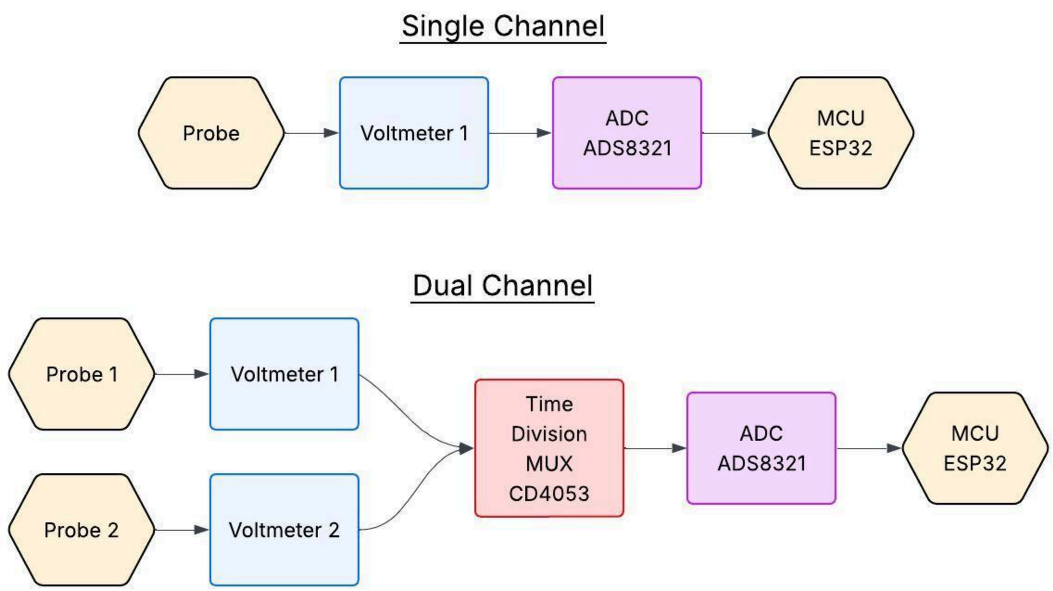
6. Oscilloscope Subsystem Report

6.1 Subsystem Introduction

The oscilloscope subsystem is designed to measure AC signals from the user's circuits. By building off the existing multimeter subsystem's voltmeter design, the oscilloscope measures signals through two voltmeters, acting as two channels for the oscilloscope. With the CD4053 mux, signals are sampled alternately which get time-division multiplexed to be decoded by the ESP32. Sharing an MCU with the power and signal generation subsystems, the oscilloscope subsystem is powered by the power supply subsystem.

6.2 Subsystem Details

Figure 39. Oscilloscope Subsystem Block Diagram



As seen in **Figure 39**, the oscilloscope utilizes two voltmeters as the channels for the oscilloscope, which get time-division multiplexed by the CD4053, passed through the ADC, then graphed using the MCU to display user-friendly data.

6.3 Subsystem Validation

Because the oscilloscope utilizes the same voltmeter design as the multimeter, the magnitude results are the same as **Table 3** in **Section 5.3.2** above. To measure sampling rate, the frequency of the signal generator is incrementally increased until significant distortion occurs.

6.4 Subsystem Conclusion

6.4.1 404 Changes

Between 403 and 404, the oscilloscope underwent design changes solely due to the fact that it uses the same components as the voltmeter. Due to the voltmeter being consolidated to improve affordability, the oscilloscope also underwent design changes. Because the functionality of the oscilloscope is heavily software based, it remained almost completely the same.

6.4.2 Conclusion
The oscilloscope design operates as intended, with the voltmeter channels measuring data close to their expected values, and the program isolating the different channels to graph the separate signal.

ECEN215 LAB KIT

Gavin Bernard
Maylun Huang
Evan Slyter

404 Final System Report

REVISION – 1
27 April 2026

404 Final System Report
FOR
ECEN215 Lab Kit

Approved By:

Gavin Bernard	4/27/26
Project Leader	Date

Prof. Jang	Date

Hailey Ward	4/27/26
T/A	Date

Change Record

Rev.	Date	Originator	Approvals	Description
1	4/27/2026	ECEN215 Lab Kit		Initial full release

Table of Contents

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1. Introduction

The ECEN 215 Labkit is designed specifically for students to be able to complete every lab individually through an app. The app (Evan Slyter) walks students through each step of each lab. As well, it gives diagrams of how each lab circuit should be designed to get the correct output. To end the labs on the app, it gives specific instructions on which measurements to take and then exports those values to the students laptop to then build their lab report. The physical hardware within the Labkit has a DC Power Supply and Signal Generator (Gavin Bernard). The DC Power Supply, powers the entire Labkit with very little voltage sag. As well, it has multiple different voltage outputs that can power the circuits the students build for the lab. The Signal Generator is used as an input as well for the specific labs that required an AC signal. Finally, to take measurements with the Labkit, a multimeter and oscilloscope (Maylun Huang) was designed to take different measurements that are required for specific labs. Once these measurements were taken, they were scaled for the ESP32 to be able to read them accurately and then exported to the students laptop. Overall, the ECEN215 Labkit is designed for a student to be able to remotely complete any lab from home with a wall outlet and a phone with bluetooth.

2. Hardware Integration

2.1 Introduction

The hardware was designed to be compact and compatible with an ESP32. The DC power supply was designed within the PCB to take in a 12-volt power supply and power the entire board while also powering the labs the students design. Every part of the PCB was designed around the 3.3 volt rail, -5 volt rail, and 5 volt rail. As well the PCB was designed to be connected by bluetooth to the students phone. The ESP32 was connected through the TX and RX pins as well as the GND and Vin pins to upload code via USB-to-UART connection.

2.2 PCB Integration

2.2.1 PCB Limitations

The PCB needed to be designed as small as possible for the system to be shipped to students. As well, it needed to be light as this would allow for the cheapest shipping possible. While considering the size, the system also needed to be able to connect to the students breadboard from all points.

2.2.2 PCB Edits

The PCB design originally started from the first iteration of each subsystem in ECEN 403. Changes had to be made to the buck converters as the current load out of the specific IC's wasn't able to support the entire system. In addition, the digital potentiometer from the original design wasn't accurate enough with the ESP32 to be able to create a confident input voltage for the labs. This was switched with a DAC that doesn't supply a large load current, but gives a very accurate voltage output. As well, the signal generator was originally designed to be powered by the 5 volt rail, but the entire system was later switched to be powered by the 3.3 volt rail as the 5 volt rail was too much for the AD9833 to handle and destroyed the chip multiple times.

In order to combine the power supply subsystem with the measurement functions, the PCB increased from 2 layers to 4 layers, which allowed for additional signal and power planes. In addition, the multimeter designs changed slightly between semesters, to optimize for cost and space restraints. To account for the separate boards for MCU and power connections, additional header pins were also added to the designs as connection points.

2.2.3 PCB Combination Issues

While combining the PCB, the small deviations in the voltage sag due to the current load created issues with the multimeter and oscilloscope readings. This was especially difficult with the inverter chip only having a max current load of 40 mA which caused a large sag once connected all together. To fix this issue, designing the inverter with a higher quality and more expensive design would have created a more stable rail for the system to have higher quality readings. In addition, with four subsystems on one PCB, wiring became difficult, which was satisfied by increasing the number of layers on the board.

3. ESP32 Integration – Hardware

3.1 Introduction

The ESP32 was one of the biggest issues with our combined system as only working with ESP32 Dev Board kits in ECEN 403, we struggled designing an ESP32 that allowed us to flash code to it. We had multiple iterations of the ESP32 PCB where we had one fully integrated within one PCB and one that allowed us to jump to specific pins. The ESP32 hardware involved supplying 3.3 volts to the Vin pin, opening the TX and RX pins for code to be uploaded, and combining into one ground. The ESP32 hardware is essential for our system to work as this allows for measurements to be uploaded to the students phone and for the students to control the specific inputs for their circuits.

3.2 MCU PCB

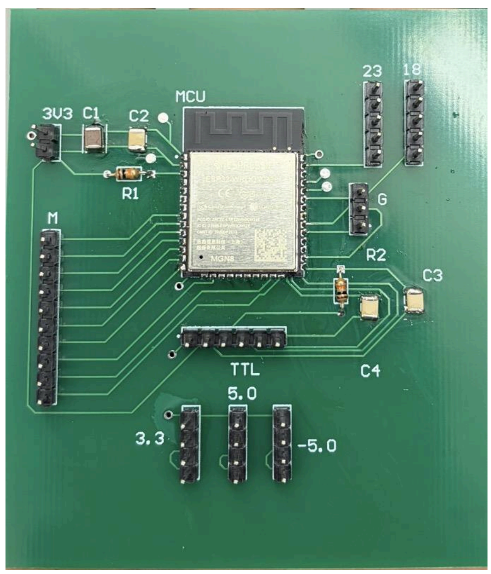
3.2.1 PCB Limitations

Within designing the PCB, the specific pins needed for each subsystem caused limitations for combining from the input subsystems and the measurement susbsytems. As well, with multiple SPI connections, the code within the MCU was very limited when trying to control multiple different SPI chips.

3.2.2 PCB Edits

For the MCU hardware, we hard originally an entirely integrated board with an ESP32 connected through a USB-to-UART connectio which would allow for the code to uploaded without an external wires. As this connection wasn't successful, we moved over to a separate MCU board that allowed for jumper wires to connect and upload code, but not being able to turn the MCU into bootload mode caused for code not to be uploaded. This issue could have been easily avoided on the main integrated board if the upload style was with the UBS-to-TTL connection instead of trying UBS-to-UART.

Figure 1: Second MCU Iteration



4. ESP32 Integration – Software

4.1 Introduction

Once the ESP32 was physically integrated (input and output ESP32s combined), the major hurdle was combining the individually working code into one larger program that would allow all subsystems to work at once without having to have the app switch between multiple devices.

4.2 BLE

The Labkit was able to connect to other bluetooth devices by advertising a BLE service. This means that the Labkit is showing other bluetooth devices that it is a device that can be connected to, instead of a device meant to connect to other services. This was done by broadcasting its name (Labkit), and its UUID (essentially a password so only authorized devices can send commands).

4.3 Data Format

Information sent from the Labkit to the mobile app is sent through a compact binary style. This is a very simple and reliable way of sending low amounts of simple numerical data. The data sent will be from the multimeter and oscilloscope subsystems. Both of these systems only send numerical values so nothing more capable of large amounts of data, such as wifi, is needed.

4.4 Combination Problems

4.4.1 Combined SPI Connection

When combining multiple chips such as the MCP4921, AD9833, and the MCP4161 the SPI bus was having a hard time allowing all chips working simultaneously. When sharing pins 23 and 18 of the MCU from all three chips and uploading code to the MCU, the app could only control the DAC and not the signal generator.

Fixing this problem would be best if two chips were using I^2C for connection to ESP32 as this would allow them to operate in isolation while the AD9833 could have the SPI pins directly to itself. The problem made integration hard from the DC power supply and the signal generator, but did not affect the multimeter or oscilloscope.

5. Mobile App Integration – Software

5.1 Introduction

Connecting the mobile app to the other subsystems was the biggest challenge in our integration process. It involved using Bluetooth Low Energy (BLE) to connect to the BLE service being advertised by the ESP32.

5.2 App Edits

The mobile app was edited extremely often during the process of integrating the other subsystems. Basic connectivity was established quickly, but as the other subsystems became more complete, and we were able to more accurately predict where the system would reasonably be by the end of the class, many edits were made.

5.3 Power Supply Integration

The power supply was the first subsystem to be integrated with the mobile app. The mobile app sends UTF-8 encoded ASCII commands through BLE to the ESP32. The first command made was to control the variable voltage of the power supply. This is done by sending a command such as V=5, to set the output voltage to any particular value within the range of 0-5 volts.

5.4 Signal Generator Integration

The signal generator was the second subsystem to be integrated with the mobile app. While the signal generator was never fully completed before our demo, the commands sent from the app to the Labkit were fully implemented. Similarly to voltage, sending an ASCII message such as f=1000 will set the frequency of the signal being generated within the range of 0 - 1500 Hz.

5.5 Multimeter and Oscilloscope Integration

The multimeter and oscilloscope were integrated into the mobile app together. The average value detected by the multimeter over a second was sent from the Labkit to the mobile app. Depending on the selection of the multimeter (voltage, resistance, current, etc), the ESP32 sends the detected voltage values, and decodes the current and resistance values. As the oscilloscope uses the same components as the voltmeter, the sampling aspect remains the same as the voltmeter, except the software graphs all the data points as opposed to averaging them.

5.7 Data Format

The data sent from the mobile app to the Labkit is sent through UTF-8 encoded ASCII commands. This is sufficient for our purposes since only very simple commands are sent from the app. The only types of commands sent are setting outputs to specific levels within a specific range.

6. Integrated System Results

6.1 Introduction

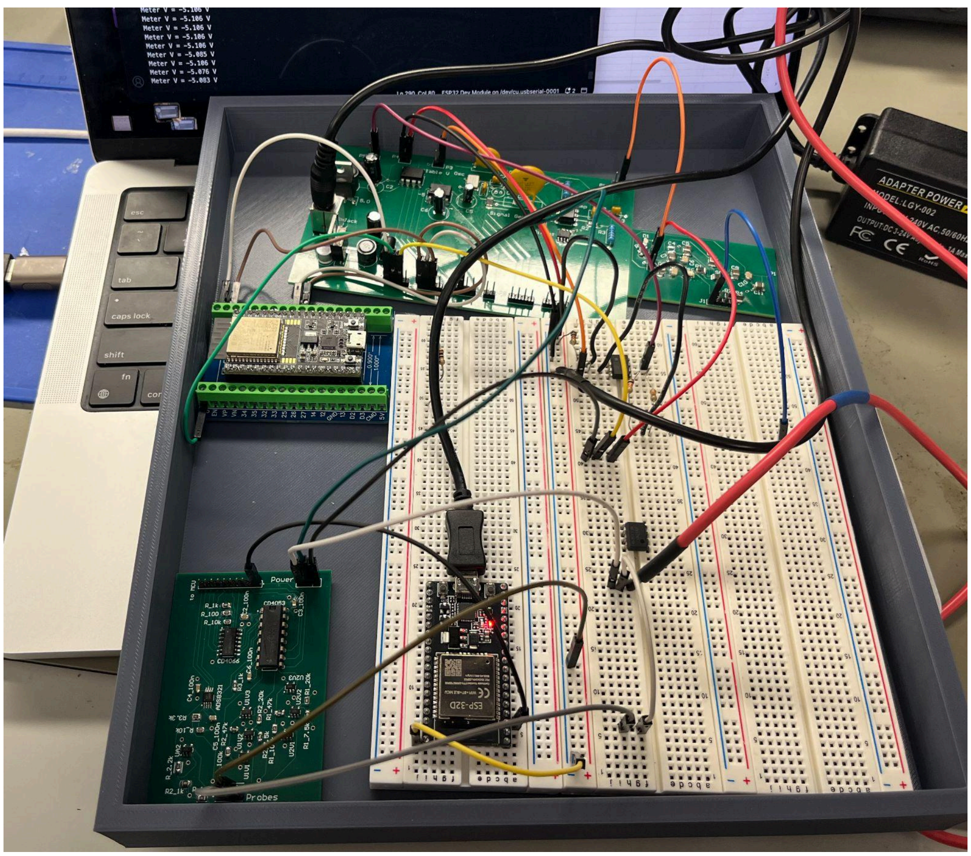
After integrating each the hardware and software subsystems, system-level validation was performed on the device to determine operating specifications and ranges. Testing of the system included both range and limit testing, as well as performing the individual labs in order to confirm the device was capable of the requirements.

6.2 Validation Testing Setup

In order to perform limit testing, the power supply, signal generator, measurements, and microcontroller board were first jumped together using the designated header pins, then connected to a separate bread board to test each capability of the multimeter. In addition to testing the expected range of each function, supplying voltage, currents, and resistances outside of the targets determine the limits of the device. The oscilloscope was tested similarly, increasing the frequency until it could no longer accurately measure the signal. The power supply testing included using the maximum load in all other subsystems in order to test how much the power supply could handle. The signal generator testing included outputting signals, and using a bench oscilloscope to validate the signal.

Performing each individual lab included setting up each lab to the lab manuals specifications, then using each function of the entire system to ensure accuracy. The setup as seen in **Figure 2** demonstrates Lab 3 on the middle breadboard, connection to the MCU, and jumps between all three of the PCBs.

Figure 2. Full System Testing



6.3 Latency

The latency of our full integrated system comes down to two parts. Firstly, how quickly the app can connect to the Labkit through bluetooth. Secondly, how quickly data can be sent back and forth between the Labkit and the app.

6.3.1 Connectivity Latency

12 trials were conducted to test the range and time to connect between the Labkit and the app. Through testing it has been determined that the Labkit can connect to the app within 5 seconds as long as the Labkit and mobile device with the app are within 6ft of each other. As the Labkit will be physically connected to a breadboard through wires when using the multimeter and power generation, 6ft was determined to be more distance than a user will ever be using the Labkit in a realistic scenario.

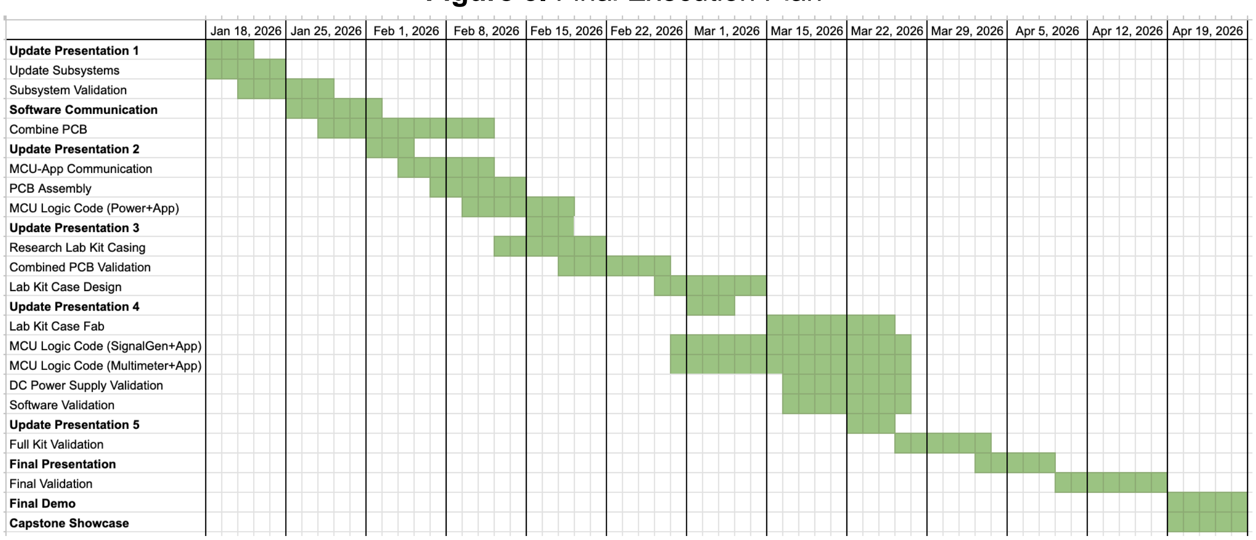
6.3.2 Labkit to App Latency

12 trials were conducted each to test the latency of data sent from the Labkit to the mobile app, and the latency of the commands sent from the app to the Labkit. In both cases, whenever the Labkit was within 4ft, the data was sent and received more quickly than I was able to start and stop a stopwatch. Within 6ft, the data was sent and received within 5 seconds if the phone was level with the bottom of the Labkit, and within 0.5 seconds if the mobile device was elevated above the Labkit.

6.5 Final Execution & Validation Plan

The final execution plan seen in **Figure 3** outlines the process of subsystem validation to full system integration, validation, and delivery. Key milestones include integrated validation for both hardware and software, sponsor updates, and refinement based on past test results. The schedule also incorporates buffer periods to account for unforeseen issues, ensuring that critical deadlines are met.

Figure 3. Final Execution Plan



The validation plan as seen in **Figure 4** below defines a systematic approach to verifying that both the subsystems and fully integrated design meet the required performance specifications. Individual subsystems are tested against quantitative metrics such as accuracy, stability, response time, and communication reliability. Following subsystem validation, integration testing ensures data consistency and overall system functionality under realistic operating conditions. This structured validation process ensures that the final system performs reliably, meets design objectives, and provides a consistent user experience.

Figure 4. Final Validation Plan

Test Name	Success Criteria	Methodology	Status	Owner
Communication Range	Labkit connects to User's device within 8 ft	Connecting device to labkit at 2ft intervals starting with device on top of labkit.	VALIDATED	Evan
Outgoing Communication Time	Labkit sends information to device in <0.5s	A stopwatch will be started at the same time as the Labkit sends data to the phone. Once data is being recieved, the timer will be stopped.	UNTESTED	Evan
Connection Time	Labkit connects to users device within 30 seconds of scanning	A stopwatch will be started as soon as the device begins scanning for a bluetooth connection. Once the scan finds the Labkit and the Labkit succesfully connects, the timer will be stopped	VALIDATED	Evan
Communication Accuracy	Binary data recieved on Labkit is the same data sent from User's device	Data is sent from the User's device to the labkit. We will ensure we know which sequence is being sent and check what arrives at the MCU.	VALIDATED	Evan
Communication Strength	Labkit connects to User's device through minor obstacles.	Connecting the device to the labkit when the labkit is covered by a blanket, under a desk, and behind a door.	VALIDATED	Evan
Incoming Communication Time	Labkit recieves information from device in <0.75s	A stopwatch will be started at the same time as the phone sends a command to the Labkit. Once the command is received and completed, the timer will be stopped.	VALIDATED	Evan
Power supplies correct voltage following command from app	Power supply sucessfully sets output voltage to whatever is said by binary data.	Once binary accuracy is confirmed, see if MCU code reads it correctly and sets output voltage as expected for voltages of 0,0.5,3.3, and 5.	VALIDATED	Evan + Gavin
Signal Generator can be controlled by the App	Signal Generator successfully changes the frequency,waveform, and voltage to whatever is inputted from the app.	Input from the app using code that was developed prior and then test using an oscilliscope if the output is correct to the expected input.	PARTIALLY VALIDATED	Evan + Gavin
Multimeter Voltage Testing	Measures -5 to 5V within 2% accuracy	Use bench multimeter to measure scaled voltage measurements, and compare to ideal measurements, then decode using MCU code	VALIDATED	Maylun
Multimeter Current Testing	Measures 0 to 1A within 2% accuracy	Use bench multimeter to measure scaled voltage measurements, and compare to ideal measurements, then decode using MCU code	VALIDATED	Maylun
Multimeter Resistance Testing	Measures 10 to 10kΩ within 2% accuracy	Use bench multimeter to measure scaled voltage measurements, and compare to ideal measurements, then decode using MCU code	VALIDATED	Maylun
Oscilloscope	Graph signals up to 10kHz	Use bench oscilloscope to graph outputted signal, and use MCU to graph output for single and dual channel	VALIDATED	Maylun
LabKit Measurements	LabKit receives and formats measurements	Apply multimeter and oscilloscope, and view output on Mobile App	PARTIALLY VALIDATED	Maylun
LabKit 5 Volt Output	LabKit is able to output 5 Volts and 1 Amps	12 Volt and 5 amp input uses a buck converter to step down the voltage and current to 5 volts and 1 amps	VALIDATED	Gavin
LabKit 3.3 Volt Output	LabKit is able to output 3.3 Volts and 1 Amp	Takes input and steps down the voltage to 3.3 volts and 1 amp to help power the ESP32 and other components within the system	VALIDATED	Gavin
LabKit Controllable Variable Voltage	LabKit is able to control a variable voltage output from 0 to 5 Volts	Uses a digital-to-analog converter that is connected to the ESP 32 that allows for a controllable output voltage that can be used for different aspects in the circuits	VALIDATED	Gavin
LabKit -5 Volt Output	LabKit is able to output -5 Volts with 40 mA	Uses an inverter to change the 5 volt input to -5 volts so students can operate op amps	VALIDATED	Gavin
Sine, Ramp, and Square Signal Generator	Able to generate a Sine, Ramp, and Square Signal	Signal generator is connected to the ESP 32 that allows for a controllable waveform	PARTIALLY VALIDATED	Gavin
Signal Generation	Able to output a signal that can be measured	A waveform is able to be outputted and connected to a circuit for students to measure	VALIDATED	Gavin
20 - 1,500 Hz Frequency Change	Able to change the signal frequency in the range of 20 Hz to 1,500 Hz	With the Signal generator connected to the ESP 32, it can also change the frequency within the range expressed	VALIDATED	Gavin
DC Power Supply Current Load	Able to power the entire product and power every lab needed.	While fully integrated, able to support every lab with enough current for students to take measurements.	VALIDATED	Gavin
System Reset Time	Labkit becomes availble to connect to users device within 30 seconds of being powered	A stopwatch will be started as the Labkit is connected to power. The stopwatch will be stopped once the Labkit is detected by the user's device	VALIDATED	Evan
0.5 - 2.0 V Op Amp Controllable Output	Able to control the signal voltage from 0.5 Volts to 2 Volts	External op amp connected to a digital potentiometer can be regulated by the ESP 32 to help change the voltage of the signal created	PARTIALLY VALIDATED	Gavin

6.6 Overall Conclusions

In ECEN 404, the Lab Kit was taken from completely separate subsystems into a complete and integrated system, capable of nearly all functional requirements. Though not all of the validation plan has been fully verified, it is expected that an additional month could provide the necessary time frame for the complete functionality. In addition, due to the unforeseen microcontroller soldering issues, ordering separate PCBs for each of the different LabKit functions was the best solution. Adjusting the MCU design on its own PCB allowed the isolation and diagnosis of the issue, though additional time would allow for the combination of the PCB. Despite this, the integration and validation satisfy the basic requirements of the ECEN215 course, meaning the capstone project was successful.